CHAPTER 21

Carboxylic Acid Derivatives: Nucleophilic Acyl Substitution Reactions



FIGURE 21.1 **The lives of rock climbers depend on their ropes, typically made of a nylon polymer prepared by a nucleophilic acyl substitution reaction.** (credit: modification of work "110823-A-NR754-001" by USASOC News Service/Flickr, CC BY 2.0)

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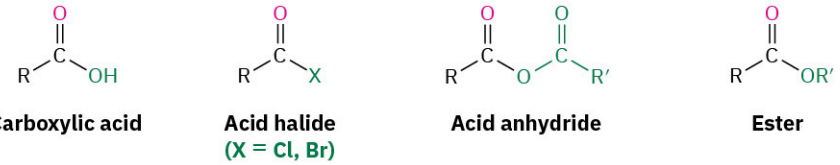
WHY THIS CHAPTER? Carboxylic acid derivatives are among the most widely occurring of all molecules, both in laboratory chemistry and in biological pathways. Thus, a study of them and their primary reaction—nucleophilic acyl substitution—is fundamental to understanding organic chemistry. We'll begin this chapter by first learning about carboxylic acid derivatives, and we'll then explore the chemistry of acyl substitution reactions.

Closely related to the carboxylic acids and nitriles discussed in the previous chapter are the **carboxylic acid**

derivatives, compounds in which an acyl group is bonded to an electronegative atom or substituent that can act as a leaving group in the nucleophilic acyl substitution reaction that we saw briefly in the **Preview of Carbonyl Chemistry**:



Many kinds of acid derivatives are known, but we'll be concerned primarily with four of the most common ones: **acid halides**, **acid anhydrides**, **esters**, and **amides**. Acid halides and acid anhydrides are used only in the laboratory, while esters and amides are common in both laboratory and biological chemistry. In addition, carboxylic acid derivatives called **thioesters** and **acyl phosphates** are encountered primarily in biological chemistry. Note the structural similarity between acid anhydrides and acyl phosphates.



Carboxylic acid | **Acid halide (X = Cl, Br)** | **Acid anhydride** | **Ester**

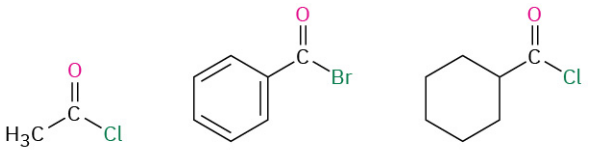


Amide | **Thioester** | **Acyl phosphate**

21.1 Naming Carboxylic Acid Derivatives

Acid Halides, RCOX

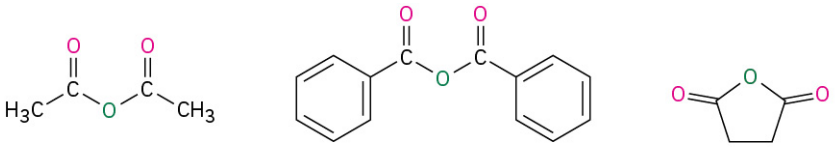
Acid halides are named by identifying first the acyl group and then the halide. As described in **Section 20.1** and shown in **TABLE 20.1**, the acyl group name is derived from the carboxylic acid name by replacing the *-ic acid* or *-oic acid* ending with *-oyl,* or the *-carboxylic acid* ending with *-carbonyl.* To keep things interesting, however, IUPAC recognizes eight exceptions for which a *-yl* rather than an *-oyl* ending is used: formic (formyl), acetic (acetyl), propionic (propionyl), butyric (butyryl), oxalic (oxalyl), malonic (malonyl), succinic (succinyl), and glutaric (glutaryl).



Acetyl chloride | **Benzoyl bromide** | **Cyclohexanecarbonyl chloride**

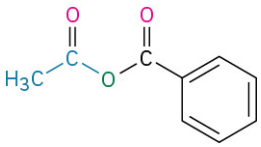
Acid Anhydrides, RCO_2COR'

Symmetrical anhydrides of unsubstituted monocarboxylic acids and cyclic anhydrides of dicarboxylic acids are named by replacing the word *acid* with *anhydride.*



Acetic anhydride | **Benzoic anhydride** | **Succinic anhydride**

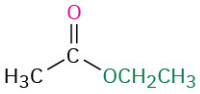
Unsymmetrical anhydrides—those prepared from two different carboxylic acids—are named by listing the two acids alphabetically and then adding *anhydride.*



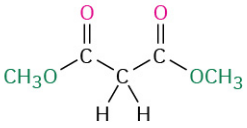
Acetic benzoic anhydride

Esters, RCO_2R'

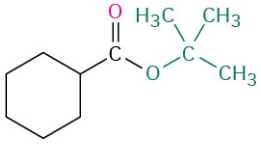
Esters are named by first identifying the alkyl group attached to oxygen and then the carboxylic acid, with the *-ic acid* ending replaced by *-ate.*



Ethyl acetate



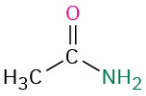
Dimethyl malonate



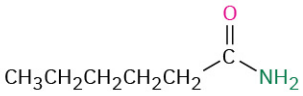
***tert*-Butyl cyclohexane-carboxylate**

Amides, $RCONH_2$

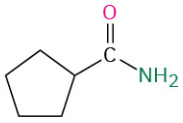
Amides with an unsubstituted $–NH_2$ group are named by replacing the *-oic acid* or *-ic acid* ending with *-amide,* or by replacing the *-carboxylic acid* ending with *-carboxamide.*



Acetamide

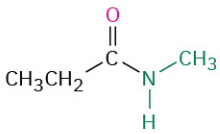


Hexanamide

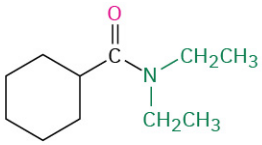


Cyclopentane-carboxamide

If the nitrogen atom is further substituted, the compound is named by first identifying the substituent groups and then the parent amide. The substituents are preceded by the letter *N* to identify them as being directly attached to nitrogen.



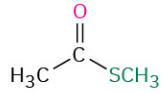
***N*-Methylpropanamide**



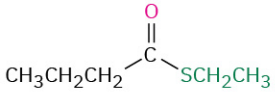
***N,N*-Diethylcyclohexanecarboxamide**

Thioesters, $RCOSR'$

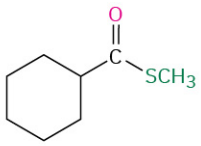
Thioesters are named like the corresponding esters. If the related ester has a common name, the prefix *thio-* is added to the name of the carboxylate: acetate becomes thioacetate, for instance. If the related ester has a systematic name, the *-oate* or *-carboxylate* ending is replaced by *-thioate* or *-carbothioate:* butanoate becomes butanethioate and cyclohexanecarboxylate becomes cyclohexanecarbothioate, for instance.



Methyl thioacetate



Ethyl butanethioate

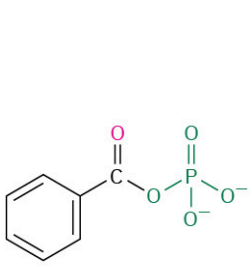


Methyl cyclohexane-carbothioate

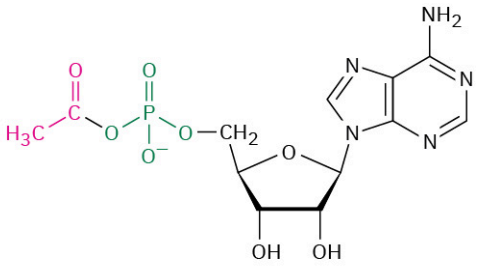
Acyl Phosphates, $RCO_2PO_3^{2-}$ and $RCO_2PO_3R'^-$

Acyl phosphates are named by citing the acyl group and adding the word *phosphate.* If an alkyl group is attached to one of the phosphate oxygens, it is identified after the name of the acyl group. In biological chemistry, acyl

adenosyl phosphates are particularly common.



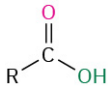
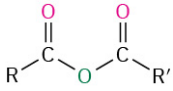
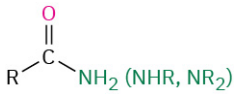
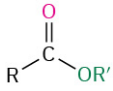
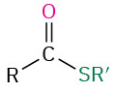
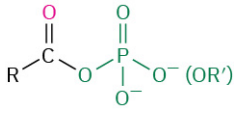
Benzoyl phosphate



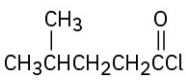
Acetyl adenosyl phosphate

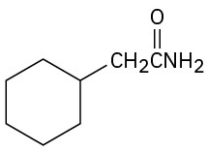
A summary of nomenclature rules for carboxylic acid derivatives is given in TABLE 21.1.

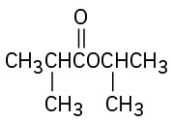
TABLE 21.1 Nomenclature of Carboxylic Acid Derivatives

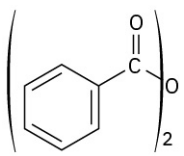
Functional group	Structure	Name ending
Carboxylic acid		*-ic acid (-carboxylic acid)*
Acid halide		*-oyl halide (-carbonyl halide)*
Acid anhydride		*anhydride*
Amide		*-amide (-carboxamide)*
Ester		*-oate (-carboxylate)*
Thioester		*-thioate (-carbothioate)*
Acyl phosphate		*-oyl phosphate*

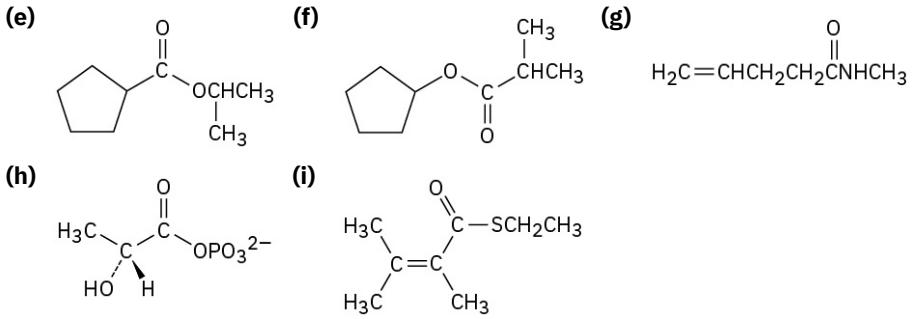
PROBLEM 21-1 Give IUPAC names for the following substances:

(a) 

(b) 

(c) 

(d) 



PROBLEM Draw structures corresponding to the following names:

21-2 **(a)** Phenyl benzoate **(b)** *N*-Ethyl-*N*-methylbutanamide **(c)** 2,4-Dimethylpentanoyl chloride
(d) Methyl 1-methylcyclohexanecarboxylate **(e)** Ethyl 3-oxopentanoate
(f) Methyl *p*-bromobenzenethioate **(g)** Formic propanoic anhydride
(h) *cis*-2-Methylcyclopentanecarbonyl bromide

21.2 Nucleophilic Acyl Substitution Reactions

The addition of a nucleophile to a polar C=O bond is the key step in three of the four major carbonyl-group reactions. We saw in the chapter on **Aldehydes and Ketones: Nucleophilic Addition Reactions** that when a nucleophile adds to an aldehyde or ketone, the initially formed tetrahedral intermediate can be protonated to yield an alcohol. When a nucleophile adds to a carboxylic acid derivative, however, a different reaction path is taken. The initially formed tetrahedral intermediate eliminates one of the two substituents originally bonded to the carbonyl carbon, leading to a net **nucleophilic acyl substitution reaction** (FIGURE 21.2).

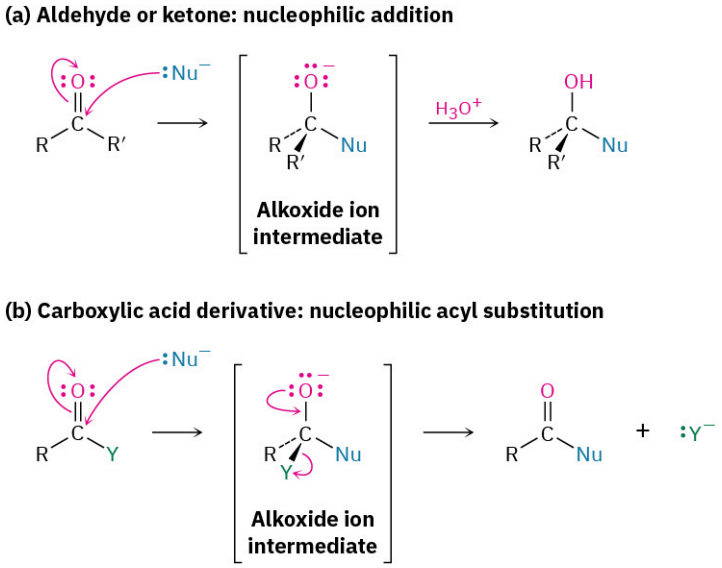
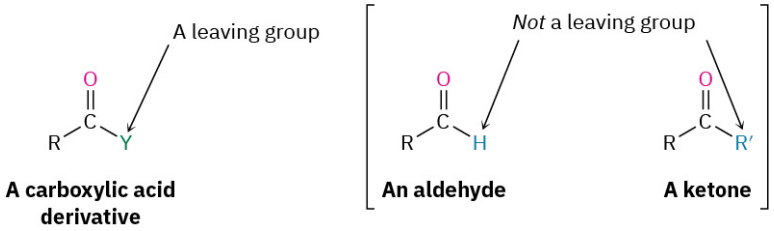


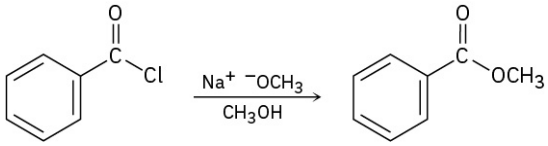
FIGURE 21.2 The general mechanisms of nucleophilic addition and nucleophilic acyl substitution reactions. Both reactions begin with addition of a nucleophile to a polar C=O bond to give a tetrahedral, alkoxide ion intermediate. **(a)** The intermediate formed from an aldehyde or ketone is protonated to give an alcohol, but **(b)** the intermediate formed from a carboxylic acid derivative expels a leaving group to give a new carbonyl compound.

The difference in behavior between aldehydes/ketones and carboxylic acid derivatives is a consequence of structure. Carboxylic acid derivatives have an acyl carbon bonded to a group –Y that can act as a leaving group, often as a stable anion. As soon as the tetrahedral intermediate is formed, the leaving group is expelled to generate a new carbonyl compound. Aldehydes and ketones, however, have no such leaving group, however, and therefore don't undergo substitution.



The net effect of the addition/elimination sequence is a substitution of the nucleophile for the –Y group that was originally bonded to the acyl carbon. Thus, the overall reaction is superficially similar to the kind of nucleophilic substitution that occurs during an S_N2 reaction (**Section 11.3**), but the mechanisms of the two reactions are completely different. An S_N2 reaction occurs in a single step by backside displacement of the leaving group, while a nucleophilic acyl substitution takes place in two steps and involves a tetrahedral intermediate.

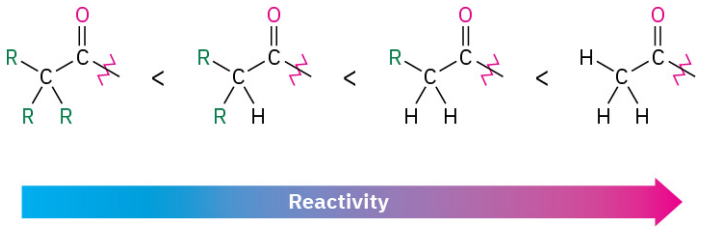
PROBLEM 21-3 Show the mechanism of the following nucleophilic acyl substitution reaction, using curved arrows to indicate the electron flow in each step:



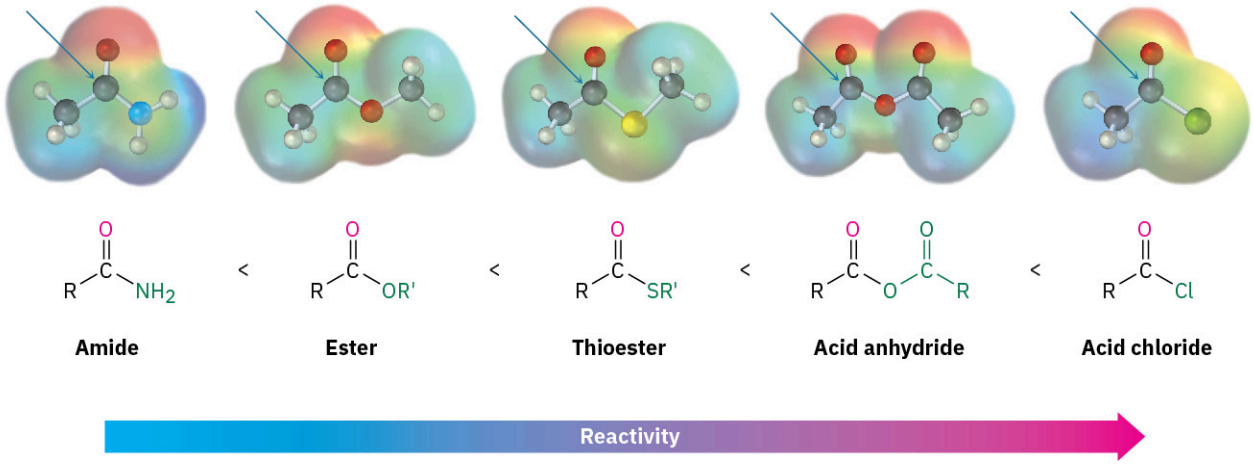
Relative Reactivity of Carboxylic Acid Derivatives

Both the initial addition step and the subsequent elimination step can affect the overall rate of a nucleophilic acyl substitution reaction, but the addition is usually the rate-limiting step. Thus, any factor that makes the carbonyl group more reactive toward nucleophiles favors the substitution process.

Steric and electronic factors are both important in determining reactivity. Sterically, we find within a series of similar acid derivatives that unhindered, accessible carbonyl groups react with nucleophiles more readily than do sterically hindered carbonyl groups. The reactivity order is



Electronically, we find that strongly polarized acyl compounds react more readily than less polar ones. Thus, acid chlorides are the most reactive because the electronegative chlorine atom withdraws electrons from the carbonyl carbon, whereas amides are the least reactive. Although subtle, electrostatic potential maps of various carboxylic acid derivatives indicate these differences by the relative blueness on the C=O carbons. Acyl phosphates are hard to place on this scale because they are not often used in the laboratory, but in biological systems they appear to be somewhat more reactive than thioesters.



The way in which various substituents affect the polarization of a carbonyl group is similar to the way they affect the reactivity of an aromatic ring toward electrophilic substitution (Section 16.4). A chlorine substituent, for example, inductively withdraws electrons from an acyl group in the same way that it withdraws electrons from and thus deactivates an aromatic ring. Similarly, amino, methoxyl, and methylthio substituents donate electrons to acyl groups by resonance in the same way that they donate electrons to, and thus activate, aromatic rings.

As a consequence of these reactivity differences, it's usually possible to convert a more reactive acid derivative into a less reactive one. Acid chlorides, for instance, can be directly converted into anhydrides, thioesters, esters, and amides, but amides can't be directly converted into esters, thioesters, anhydrides, or acid chlorides. Remembering the reactivity order is therefore a way to keep track of a large number of reactions (FIGURE 21.3). Another consequence, as noted previously, is that only acyl phosphates, thioesters, esters, and amides are commonly found in nature. Acid halides and acid anhydrides react so rapidly with water that they can't exist for long in living organisms.

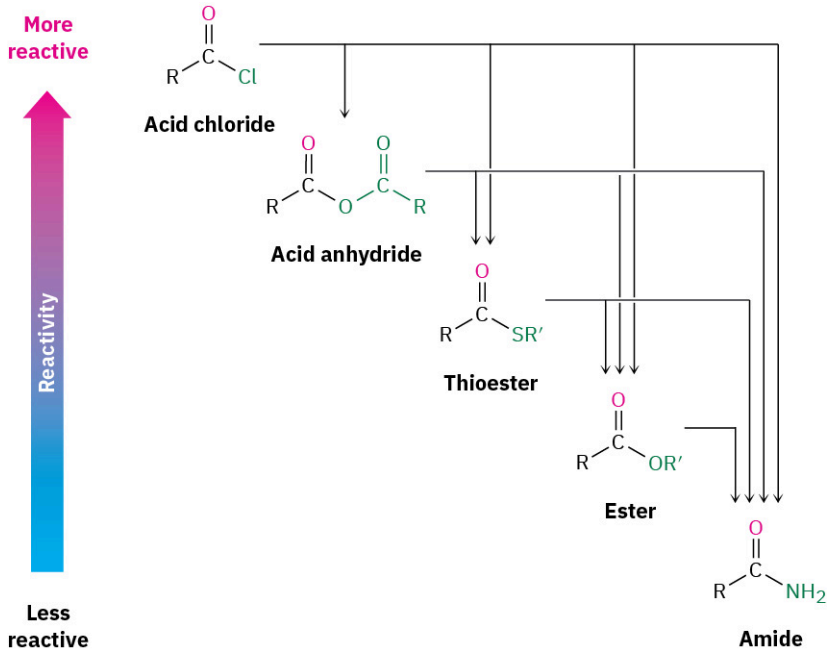


FIGURE 21.3 Interconversions of carboxylic acid derivatives. A more reactive acid derivative can be converted into a less reactive one, but not vice versa.

In studying the chemistry of carboxylic acid derivatives in the next few sections, we'll be concerned largely with the reactions of just a few nucleophiles and will see that the same kinds of reactions tend to reoccur (FIGURE 21.4).

- **Hydrolysis** Reaction with water to yield a carboxylic acid
- **Alcoholysis** Reaction with an alcohol to yield an ester
- **Aminolysis** Reaction with ammonia or an amine to yield an amide
- **Reduction** Reaction with a hydride reducing agent to yield an aldehyde or an alcohol
- **Grignard reaction** Reaction with an organometallic reagent to yield a ketone or an alcohol

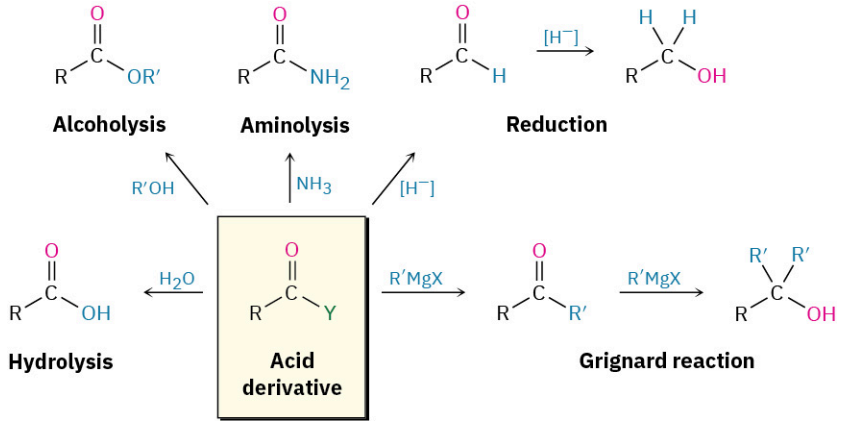
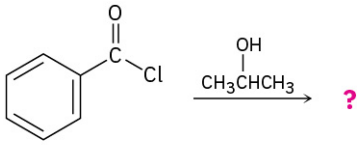


FIGURE 21.4 **Some general reactions of carboxylic acid derivatives.**

WORKED EXAMPLE 21.1

Predicting the Product of a Nucleophilic Acyl Substitution Reaction

Predict the product of the following nucleophilic acyl substitution reaction of benzoyl chloride with 2-propanol:

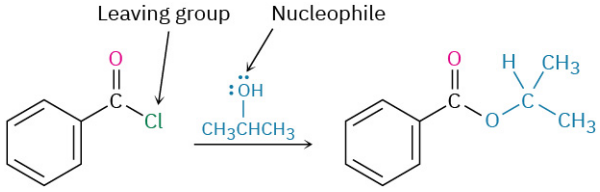


Benzoyl chloride

Strategy

A nucleophilic acyl substitution reaction involves the substitution of a nucleophile for a leaving group in a carboxylic acid derivative. Identify the leaving group (Cl^- in the case of an acid chloride) and the nucleophile (an alcohol in this case), and replace one by the other. The product is isopropyl benzoate.

Solution



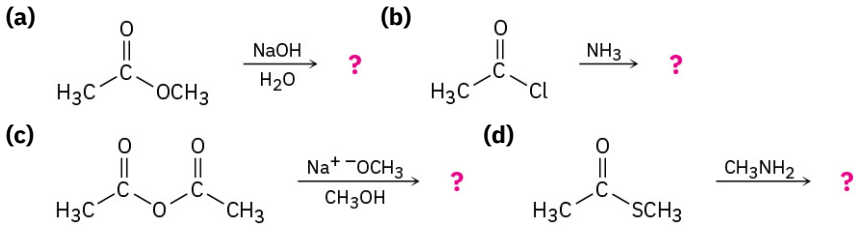
Benzoyl chloride Isopropyl benzoate

PROBLEM 21-4 Rank the compounds in each of the following sets in order of their expected reactivity toward nucleophilic acyl substitution:

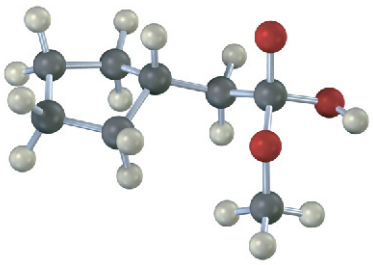
(a) CH_3CCl, CH_3COCH_3, CH_3CNH_2

(b) $CH_3COCH_2CH_3$, $CH_3COCH_2CCl_3$, $CH_3COCH(CF_3)_2$

PROBLEM 21-5 Predict the products of the following nucleophilic acyl substitution reactions:



PROBLEM 21-6 The following structure represents a tetrahedral alkoxide ion intermediate formed by addition of a nucleophile to a carboxylic acid derivative. Identify the nucleophile, the leaving group, the starting acid derivative, and the ultimate product.

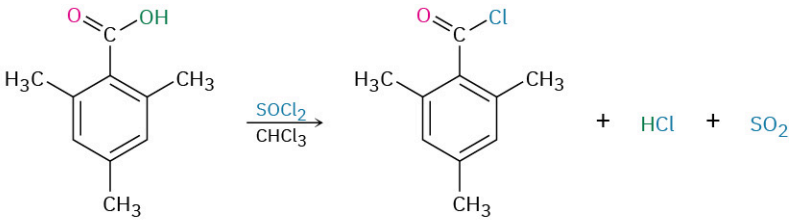


21.3 Reactions of Carboxylic Acids

The direct nucleophilic acyl substitution of a carboxylic acid is difficult because –OH is a poor leaving group (**Section 11.3**). Thus, it's usually necessary to enhance the reactivity of the acid, either by using a strong acid catalyst to protonate the carboxyl and make it a better acceptor or by converting the –OH into a better leaving group. Under the right circumstances, however, acid chlorides, anhydrides, esters, and amides can all be prepared from carboxylic acids by nucleophilic acyl substitution reactions.

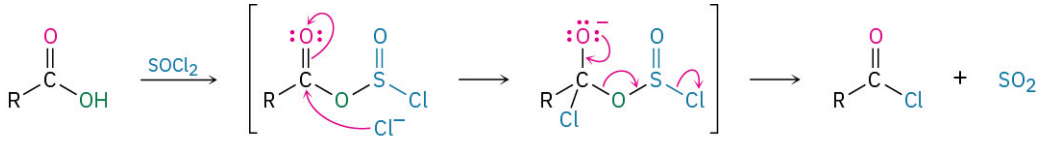
Conversion of Carboxylic Acids into Acid Chlorides

In the laboratory, carboxylic acids are converted into acid chlorides by treatment with thionyl chloride, $SOCl_2$.



2,4,6-Trimethylbenzoic acid **2,4,6-Trimethylbenzoyl chloride (90%)**

This reaction occurs by a nucleophilic acyl substitution pathway in which the carboxylic acid is first converted into an acyl chlorosulfite intermediate, thereby replacing the –OH of the acid with a much better leaving group. The chlorosulfite then reacts with a nucleophilic chloride ion. You might recall from **Section 17.6** that an analogous chlorosulfite is involved in the reaction of an alcohol with $SOCl_2$ to yield an alkyl chloride.

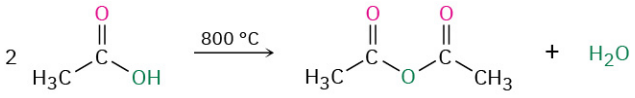


Carboxylic acid **A chlorosulfite** **Acid chloride**

Conversion of Carboxylic Acids into Acid Anhydrides

Acid anhydrides can be derived from two molecules of carboxylic acid by heating to remove 1 equivalent of water. Because of the high temperatures needed, however, only acetic anhydride is commonly prepared this

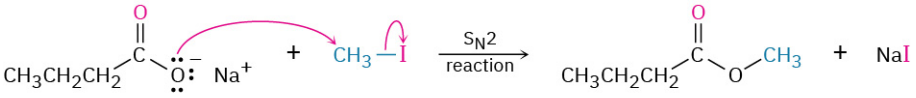
way.



Acetic acid **Acetic anhydride**

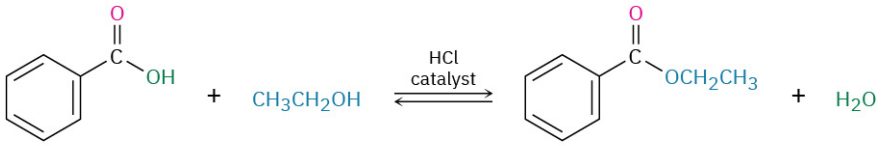
Conversion of Carboxylic Acids into Esters

Perhaps the most useful reaction of carboxylic acids is their conversion into esters. There are many methods for accomplishing this, including the S_N2 reaction of a carboxylate anion with a primary alkyl halide that we saw in **Section 11.3**.



Sodium butanoate **Methyl butanoate (97%)**

Esters can also be synthesized by an acid-catalyzed nucleophilic acyl substitution reaction of a carboxylic acid with an alcohol, a process called the **Fischer esterification reaction**. Unfortunately, the need for an excess of a liquid alcohol as solvent effectively limits the method to the synthesis of methyl, ethyl, propyl, and butyl esters.

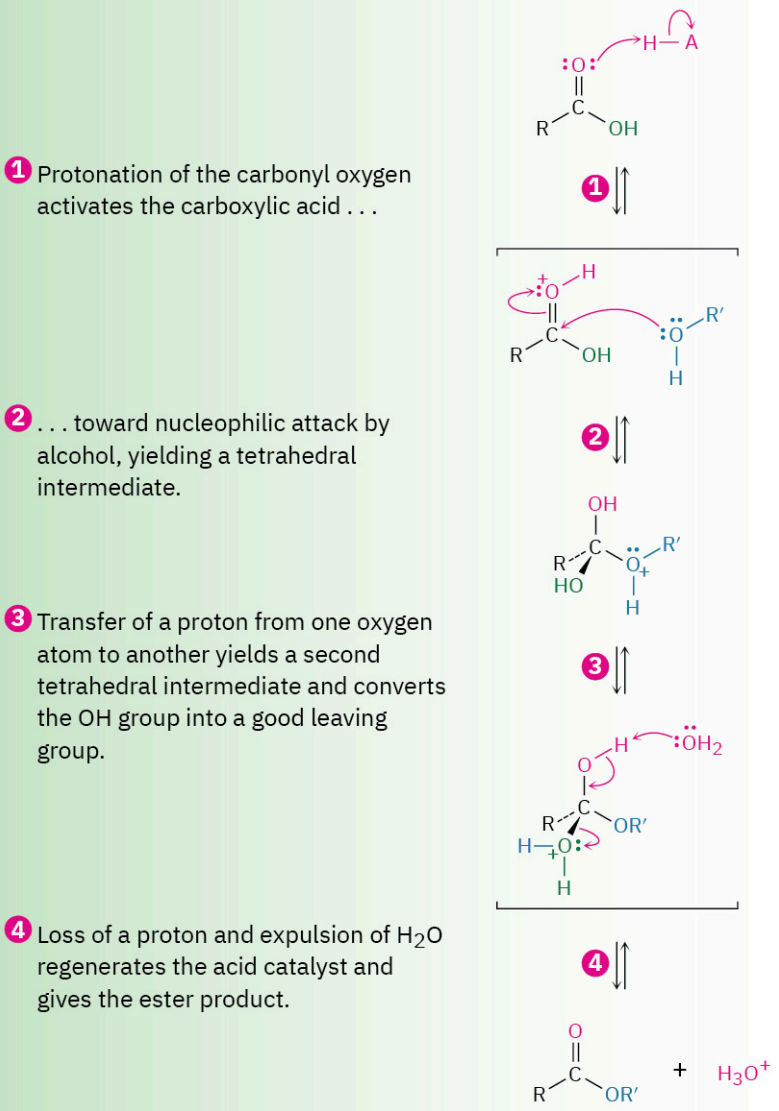


Benzoic acid **Ethyl benzoate (91%)**

The mechanism of the Fischer esterification reaction is shown in **FIGURE 21.5**. Carboxylic acids are not reactive enough to undergo nucleophilic addition directly, but their reactivity is greatly enhanced in the presence of a strong acid such as HCl or H_2SO_4. The mineral acid protonates the carbonyl-group oxygen atom, thereby giving the carboxylic acid a positive charge and rendering it much more reactive toward nucleophiles. Subsequent loss of water from the tetrahedral intermediate yields the ester product.

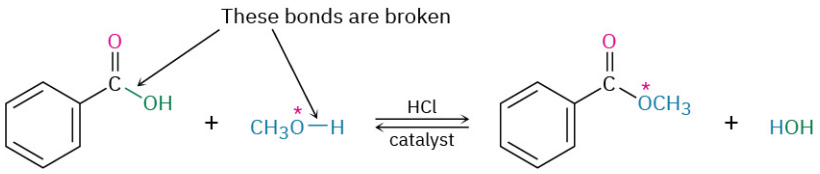
FIGURE 21.5 MECHANISM

Mechanism of Fischer esterification. The reaction is an acid-catalyzed, nucleophilic acyl substitution of a carboxylic acid.



The net effect of Fischer esterification is substitution of an –OH group by –OR′. All steps are reversible, and the reaction typically has an equilibrium constant close to 1. Thus, the reaction can be driven in either direction by the choice of reaction conditions. Ester formation is favored when a large excess of alcohol is used as solvent, but carboxylic acid formation is favored when a large excess of water is present.

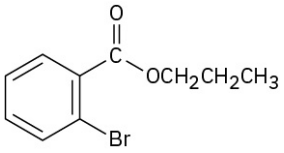
Evidence in support of the mechanism shown in FIGURE 21.5 comes from isotope-labeling experiments. When ^{18}O-labeled methanol reacts with benzoic acid, the methyl benzoate produced is found to be ^{18}O-labeled whereas the water produced is unlabeled. Thus, it is the C–OH bond of the carboxylic acid that is broken during the reaction rather than the CO–H bond and the RO–H bond of the alcohol that is broken rather than the R–OH bond.



WORKED EXAMPLE 21.2

Synthesizing an Ester from an Acid

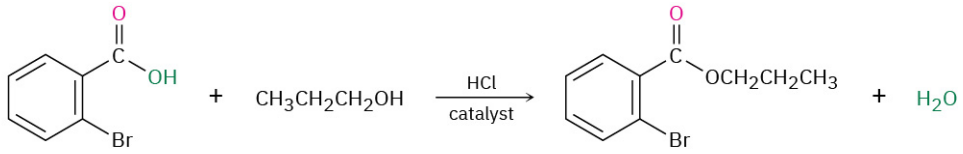
How might you prepare the following ester using a Fischer esterification reaction?



Strategy

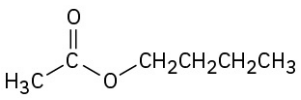
Begin by identifying the two parts of the ester. The acyl part comes from the carboxylic acid and the –OR part comes from the alcohol. In this case, the target molecule is propyl *o*-bromobenzoate, so it can be prepared by treating *o*-bromobenzoic acid with 1-propanol.

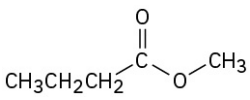
Solution

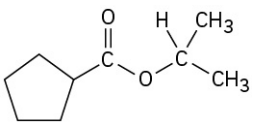


***o*-Bromobenzoic acid** **1-Propanol** **Propyl *o*-bromobenzoate**

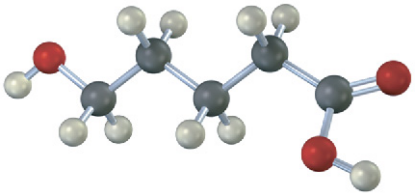
PROBLEM 21-7 How might you prepare the following esters from the corresponding acids?

(a) 

(b) 

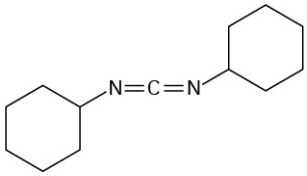
(c) 

PROBLEM 21-8 If the following molecule is treated with acid catalyst, an intramolecular esterification reaction occurs. What is the structure of the product? (*Intramolecular* means within the same molecule.)

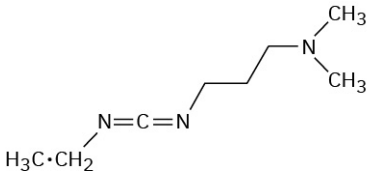


Conversion of Carboxylic Acids into Amides

Amides are difficult to prepare by direct reaction of carboxylic acids with amines because amines are bases that convert acidic carboxyl groups into their unreactive carboxylate anions. Thus, the –OH must be replaced by a better, nonacidic leaving group to carry out a nucleophilic acyl substitution. In practice, amides are often prepared by activating the carboxylic acid with a carbodiimide (R–N=C=N–R), followed by addition of the amine. Dicyclohexylcarbodiimide (DCC) and 1-ethyl-3-(3-dimethylaminopropylcarbodiimide (EDEC) are commonly used.



Dicyclohexylcarbodiimide

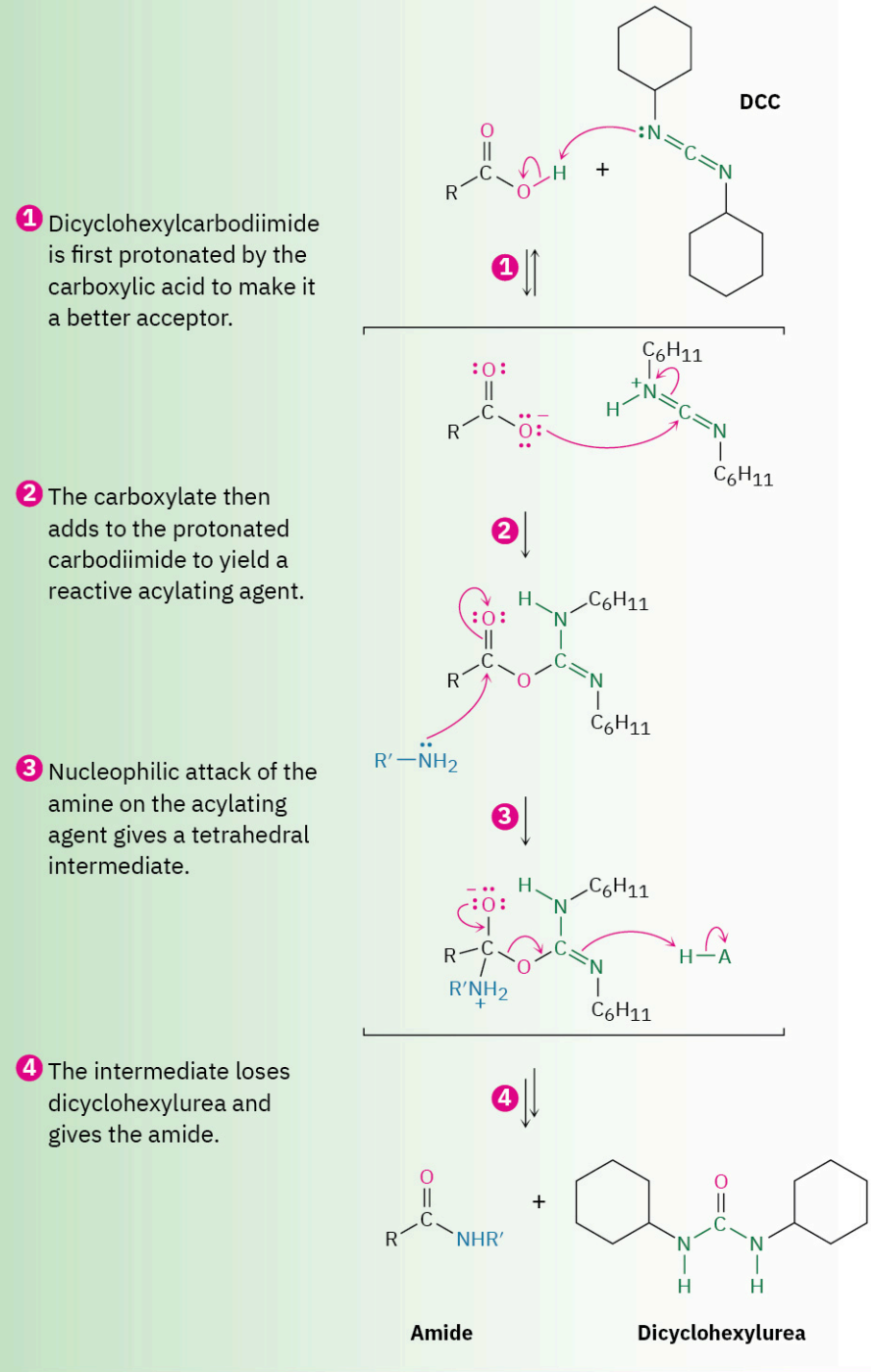


1-Ethyl-3-(3-dimethylaminopropyl)-carbodiimide

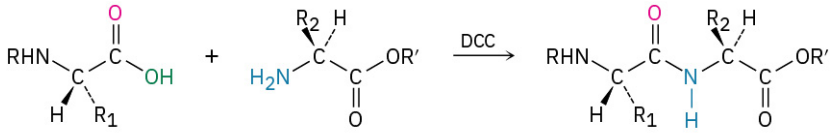
As shown in **FIGURE 21.6**, the acid first adds to a C═N double bond of DCC, and nucleophilic acyl substitution by amine then ensues. Alternatively, and depending on the reaction solvent, the reactive acyl intermediate might also react with a second equivalent of carboxylate ion to generate an acid anhydride that then reacts with the amine. The product from either pathway is the same.

FIGURE 21.6 MECHANISM

Mechanism of amide formation by reaction of a carboxylic acid and an amine with dicyclohexylcarbodiimide (DCC).



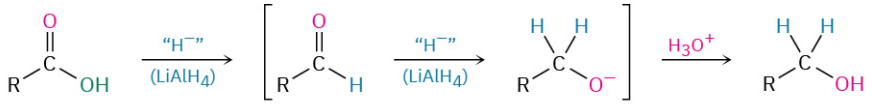
We'll see in **Section 26.7** that this carbodiimide method of amide formation is the key step in the laboratory synthesis of small proteins, or *peptides*. For instance, when one amino acid with its NH_2 rendered unreactive and a second amino acid with its $-CO_2H$ rendered unreactive are treated with carbodiimide, a dipeptide is formed.



Amino acid 1 **Amino acid 2** **A dipeptide**

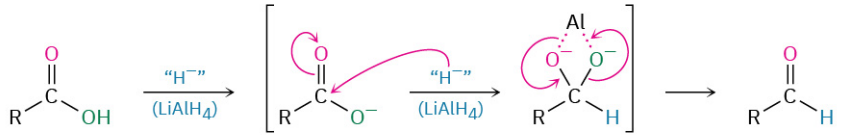
Conversion of Carboxylic Acids into Alcohols

We said in **Section 17.4** that carboxylic acids are reduced by $LiAlH_4$ to give primary alcohols, but we deferred a discussion of the reaction mechanism at that time. In fact, the reduction is a nucleophilic acyl substitution reaction in which –H replaces –OH to give an aldehyde that is further reduced by nucleophilic addition to produce a primary alcohol. The aldehyde intermediate is much more reactive than the starting acid, so it reacts instantly and is not isolated.



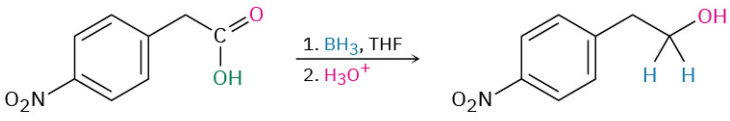
A carboxylic acid **An aldehyde (not isolated)** **An alkoxide ion** **A 1° alcohol**

Because hydride ion is a base as well as a nucleophile, the actual nucleophilic acyl substitution step takes place on the carboxylate ion rather than on the free carboxylic acid and gives a high-energy **dianion** intermediate. In this intermediate, the two oxygens are complexed to a Lewis acidic aluminum species. Thus, the reaction is relatively difficult, and acid reductions require higher temperatures and extended reaction times.



A carboxylic acid **A carboxylate** **A dianion** **An aldehyde**

Alternatively, borane in tetrahydrofuran (BH_3/THF) is a useful reagent for reducing carboxylic acids to primary alcohols. Reaction of an acid with BH_3/THF occurs rapidly at room temperature, and the procedure is often preferred to reduction with $LiAlH_4$ because of its relative ease and safety. Borane reacts with carboxylic acids faster than with any other functional group, thereby allowing selective transformations such as that on *p*-nitrophenylacetic acid. If the reduction of *p*-nitrophenylacetic acid were done with $LiAlH_4$, both the nitro and carboxyl groups would be reduced.



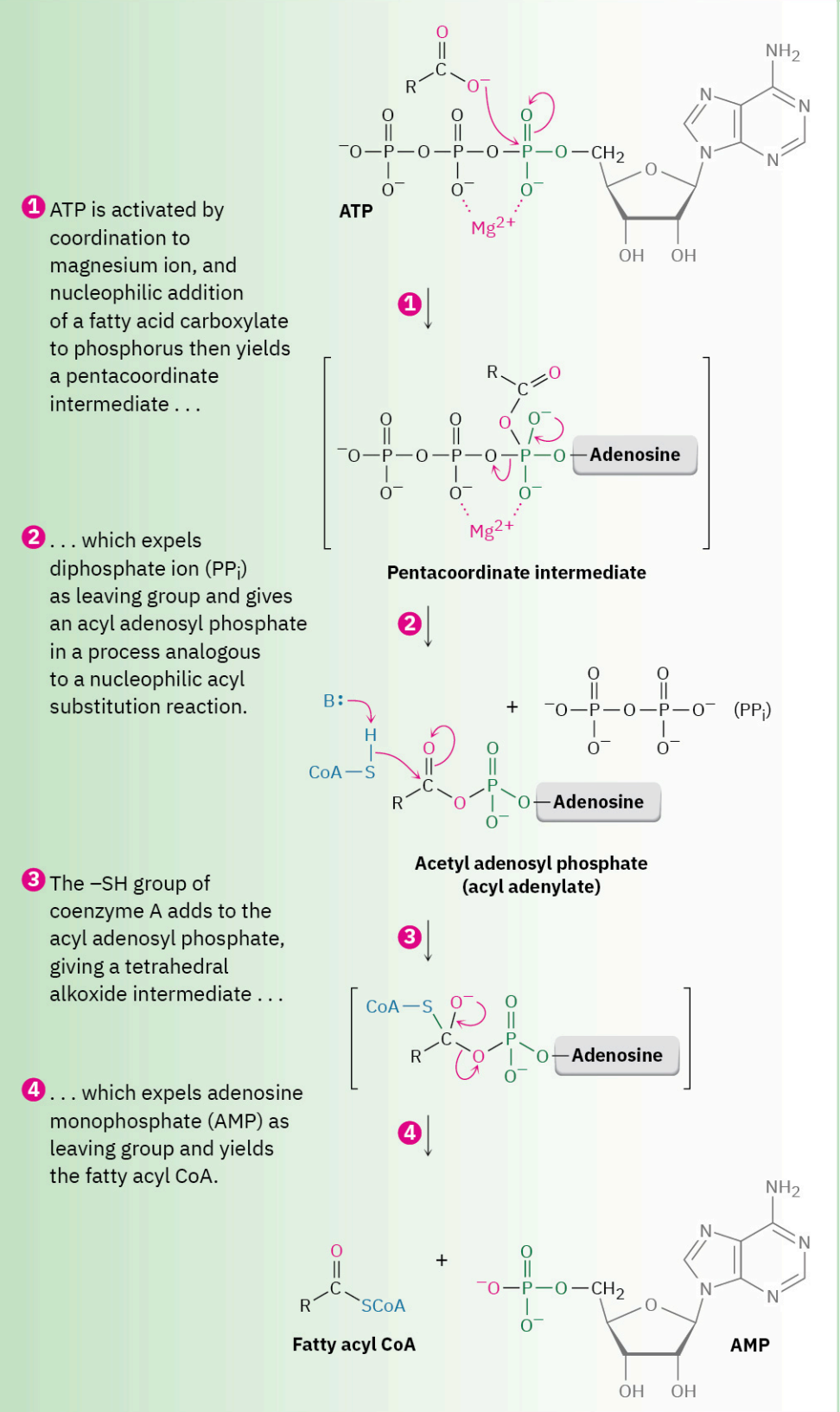
***p*-Nitrophenylacetic acid** **2-(*p*-Nitrophenyl)ethanol (94%)**

Biological Conversions of Carboxylic Acids

The direct conversion of a carboxylic acid to an acyl derivative by nucleophilic acyl substitution does not occur in biological chemistry. As in the laboratory, the acid must first be activated by converting the –OH into a better leaving group. This activation is often accomplished in living organisms by reaction of the acid with adenosine triphosphate (ATP) to give an acyl adenosyl phosphate, or *acyl adenylate,* a mixed anhydride combining a carboxylic acid and adenosine monophosphate (AMP, also known as adenylic acid). In the biosynthesis of fats, for example, a long-chain carboxylic acid reacts with ATP to give an acyl adenylate, followed by subsequent nucleophilic acyl substitution of a thiol group in coenzyme A to give the corresponding acyl CoA (**FIGURE 21.7**).

FIGURE 21.7 MECHANISM

In fatty-acid biosynthesis, a carboxylic acid is activated by reaction with ATP to give an acyl adenylate, which undergoes nucleophilic acyl substitution with the –SH group on coenzyme A. (ATP = adenosine triphosphate; AMP = adenosine monophosphate.)



Note that the first step in **FIGURE 21.7**—reaction of the carboxylate with ATP to give an acyl adenylate—is itself a nucleophilic acyl substitution *on phosphorus.* The carboxylate first adds to a P═O double bond, giving a five-coordinate phosphorus intermediate that expels diphosphate ion as a leaving group.

21.4 Chemistry of Acid Halides

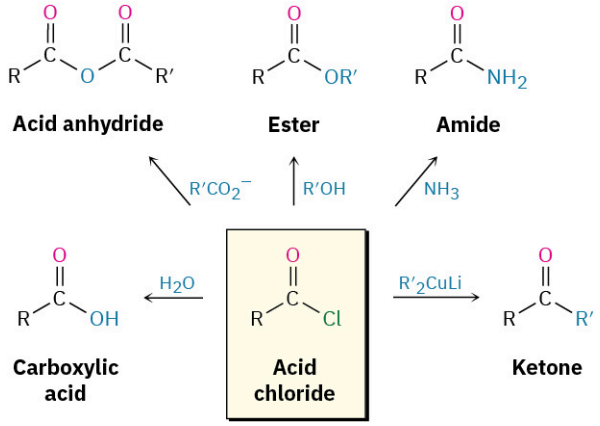
Preparation of Acid Halides

Acid chlorides are prepared from carboxylic acids by reaction with thionyl chloride ($SOCl_2$), as we saw in the previous section. Similar reaction of a carboxylic acid with phosphorus tribromide (PBr_3) yields the acid bromide.



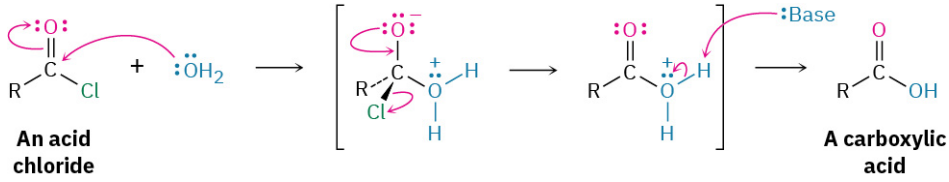
Reactions of Acid Halides

Acid halides are among the most reactive of carboxylic acid derivatives and can be converted into many other kinds of compounds by nucleophilic acyl substitution mechanisms. The halogen can be replaced by –OH to yield an acid, by –OCOR to yield an anhydride, by –OR to yield an ester, by $–NH_2$ to yield an amide, or by R′ to yield a ketone. In addition, the reduction of an acid halide yields a primary alcohol, and reaction with a Grignard reagent yields a tertiary alcohol. Although the reactions we'll be discussing in this section are illustrated only for acid chlorides, similar processes take place with other acid halides.



Conversion of Acid Halides into Carboxylic Acids: Hydrolysis

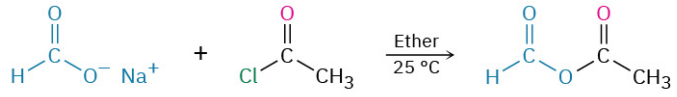
Acid chlorides react with water to yield carboxylic acids. This hydrolysis reaction is a typical nucleophilic acyl substitution process and is initiated by attack of water on the acid chloride carbonyl group. The tetrahedral intermediate undergoes elimination of Cl^- and loss of H^+ to give the product carboxylic acid plus HCl.



Because HCl is formed during hydrolysis, the reaction is often carried out in the presence of a base such as pyridine or NaOH to remove the HCl and prevent it from causing side reactions.

Conversion of Acid Halides into Anhydrides

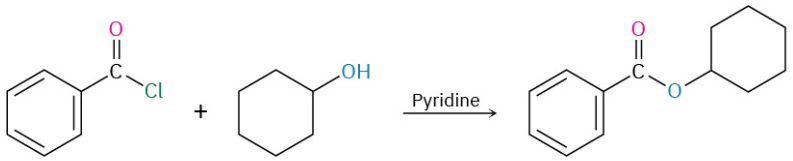
Nucleophilic acyl substitution reaction of an acid chloride with a carboxylate anion gives an acid anhydride. Both symmetrical and unsymmetrical acid anhydrides can be prepared.



Sodium formate **Acetyl chloride** **Acetic formic anhydride (64%)**

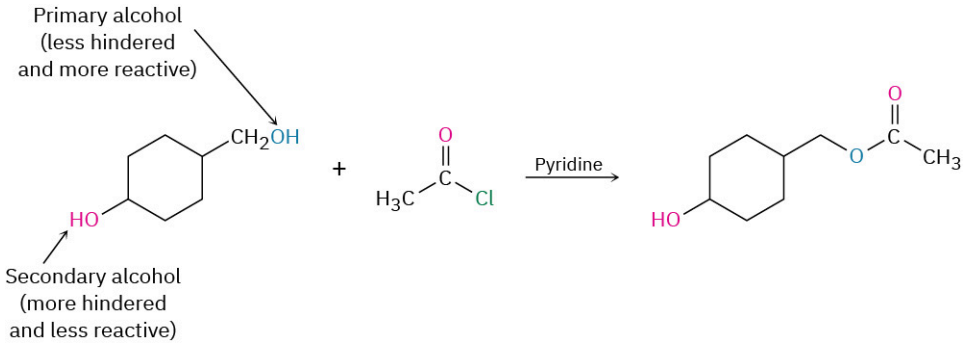
Conversion of Acid Halides into Esters: Alcoholysis

Acid chlorides react with alcohols to yield esters in a process analogous to their reaction with water to yield acids. In fact, this reaction is probably the most common method for preparing esters in the laboratory. As with hydrolysis, alcoholysis reactions are usually carried out in the presence of pyridine or NaOH to react with the HCl formed.



Benzoyl chloride **Cyclohexanol** **Cyclohexyl benzoate (97%)**

The reaction of an alcohol with an acid chloride is strongly affected by steric hindrance. Bulky groups on either partner slow down the reaction considerably, resulting in a reactivity order among alcohols of primary > secondary > tertiary. As a result, it's often possible to selectively esterify an unhindered alcohol in the presence of a more hindered one. This can be important in complex syntheses in which it's sometimes necessary to distinguish between similar functional groups. For example,



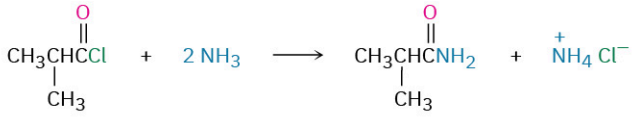
PROBLEM 21-9 How might you prepare the following esters using a nucleophilic acyl substitution reaction of an acid chloride?

(a) $CH_3CH_2CO_2CH_3$ **(b)** $CH_3CO_2CH_2CH_3$ **(c)** Ethyl benzoate

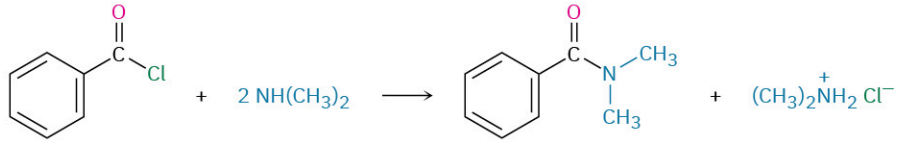
PROBLEM 21-10 Which method would you choose if you wanted to prepare cyclohexyl benzoate—Fischer esterification or reaction of an acid chloride with an alcohol? Explain.

Conversion of Acid Halides into Amides: Aminolysis

Acid chlorides react rapidly with ammonia and amines to give amides. As with the acid chloride-plus-alcohol method for preparing esters, this reaction of acid chlorides with amines is the most commonly used laboratory method for preparing amides. Both monosubstituted and disubstituted amines can be used, but not trisubstituted amines (R_3N).

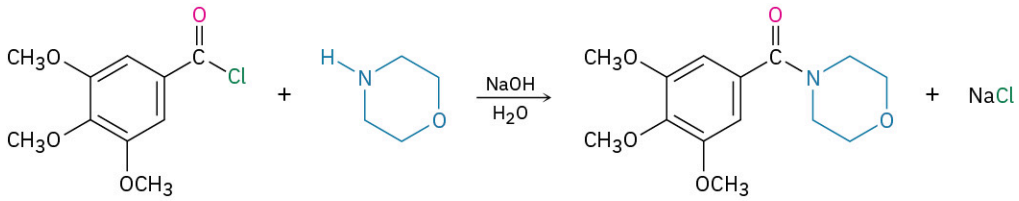


2-Methylpropanoyl chloride **2-Methylpropanamide (83%)**



Benzoyl chloride ***N,N*-Dimethylbenzamide (92%)**

Because HCl is formed during the reaction, two equivalents of the amine must be used. One equivalent reacts with the acid chloride, and one equivalent reacts with the HCl by-product to form an ammonium chloride salt. If, however, the amine component is valuable, amide synthesis is often carried out using one equivalent of the amine plus one equivalent of an inexpensive base such as NaOH. For example, the sedative trimetozine is prepared commercially by reaction of 3,4,5-trimethoxybenzoyl chloride with the amine morpholine in the presence of one equivalent of NaOH.



3,4,5-Trimethoxy-benzoyl chloride **Morpholine** **Trimetozine (an amide)**

WORKED EXAMPLE 21.3

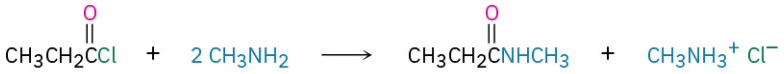
Synthesizing an Amide from an Acid Chloride

How might you prepare *N*-methylpropanamide by reaction of an acid chloride with an amine?

Strategy

As its name implies, *N*-methylpropanamide can be made by reaction of *methylamine* with the acid chloride of *propanoic* acid.

Solution



Propanoyl chloride **Methylamine** ***N*-Methylpropanamide**

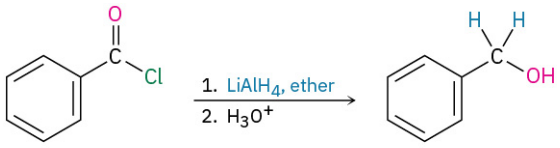
PROBLEM 21-11 Write the mechanism of the reaction just shown between 3,4,5-trimethoxybenzoyl chloride and morpholine to form trimetozine. Use curved arrows to show the electron flow in each step.

PROBLEM 21-12 How could you prepare the following amides using an acid chloride and an amine or ammonia? **(a)** $CH_3CH_2CONHCH_3$ **(b)** *N,N*-Diethylbenzamide **(c)** Propanamide

Conversion of Acid Chlorides into Alcohols: Reduction and Grignard Reaction

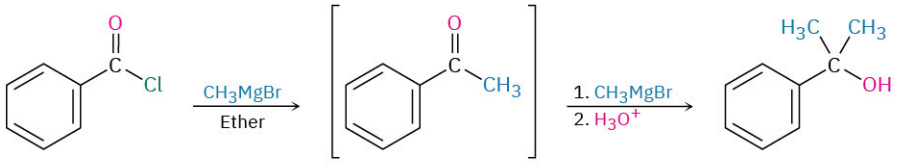
Acid chlorides are reduced by $LiAlH_4$ to yield primary alcohols. The reaction is of little practical value, however, because the parent carboxylic acids are generally more readily available and can themselves be reduced by $LiAlH_4$ to yield alcohols.

Reduction occurs via a typical nucleophilic acyl substitution mechanism in which a hydride ion ($H:^-$) adds to the carbonyl group, yielding a tetrahedral intermediate that expels Cl^-. The net effect is a substitution of –Cl by –H to yield an aldehyde, which is then further reduced by $LiAlH_4$ in a second step to yield the primary alcohol.



Benzoyl chloride **Benzyl alcohol (96%)**

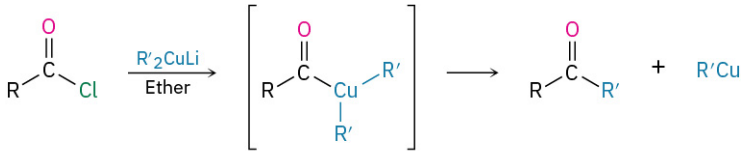
Grignard reagents react with acid chlorides to yield tertiary alcohols with two identical substituents. The mechanism of the reaction is similar to that of $LiAlH_4$ reduction. The first equivalent of Grignard reagent adds to the acid chloride, loss of Cl^- from the tetrahedral intermediate yields a ketone, and a second equivalent of Grignard reagent immediately adds to the ketone to produce an alcohol.



Benzoyl chloride **Acetophenone *(Not isolated)*** **2-Phenyl-2-propanol (92%)**

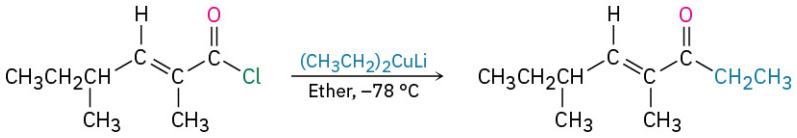
Conversion of Acid Chlorides into Ketones: Diorganocopper Reaction

The ketone intermediate formed in the reaction of an acid chloride with a Grignard reagent can't usually be isolated because addition of the second equivalent of organomagnesium reagent occurs too rapidly. A ketone *can,* however, be isolated from the reaction of an acid chloride with a lithium diorganocopper (Gilman) reagent, $Li^+ R'_2Cu^-$. The reaction occurs by initial nucleophilic acyl substitution on the acid chloride by the diorganocopper anion to yield an acyl diorganocopper intermediate, followed by loss of R′Cu and formation of the ketone.



An acid chloride **An acyl diorganocopper** **A ketone**

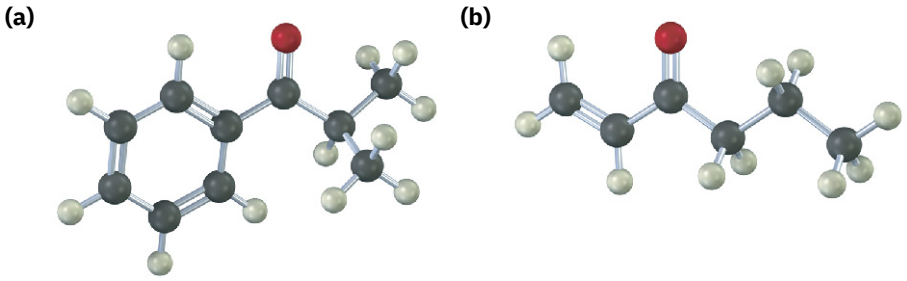
The reaction is generally carried out at –78 °C in ether solution, and yields are often excellent. For example, manicone, a substance secreted by male ants to coordinate ant pairing and mating, has been synthesized by reaction of lithium diethylcopper with (*E*)-2,4-dimethyl-2-hexenoyl chloride.



2,4-Dimethyl-2-hexenoyl chloride **4,6-Dimethyl-4-octen-3-one (manicone, 92%)**

Note that the diorganocopper reaction occurs only with acid chlorides. Carboxylic acids, esters, acid anhydrides, and amides do not react with lithium diorganocopper reagents.

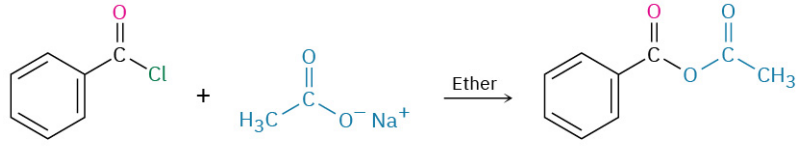
PROBLEM 21-13 How could you prepare the following ketones by reaction of an acid chloride with a lithium diorganocopper reagent?



21.5 Chemistry of Acid Anhydrides

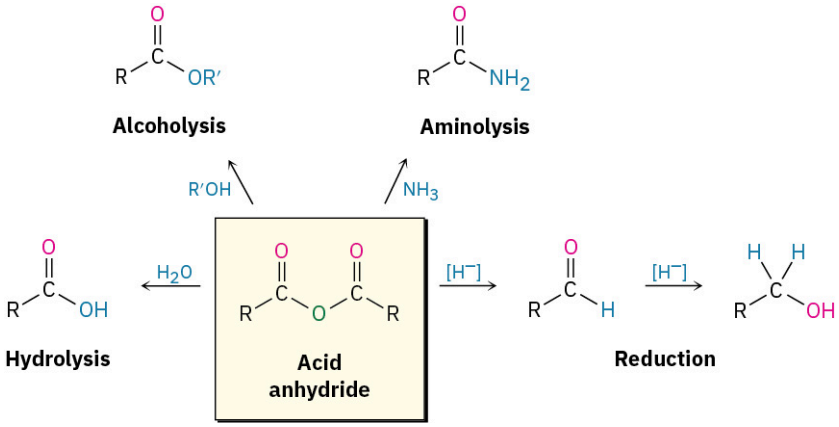
Preparation of Acid Anhydrides

Acid anhydrides are typically prepared by nucleophilic acyl substitution reaction of an acid chloride with a carboxylate anion, as we saw in the previous section. Both symmetrical and unsymmetrical acid anhydrides can be prepared in this way.



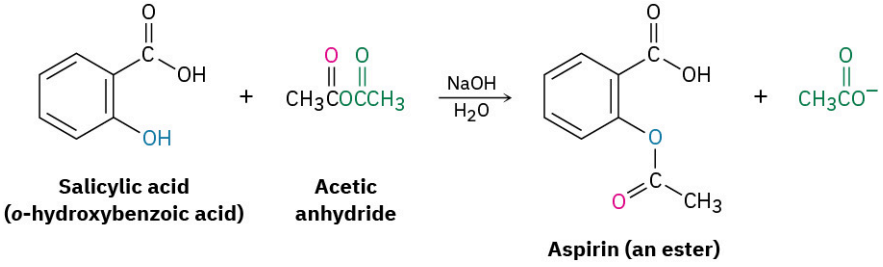
Reactions of Acid Anhydrides

The chemistry of acid anhydrides is similar to that of acid chlorides, although anhydrides react more slowly. Thus, acid anhydrides react with water to form acids, with alcohols to form esters, with amines to form amides, and with $LiAlH_4$ to form primary alcohols. Only the ester- and amide-forming reactions are commonly used, however.



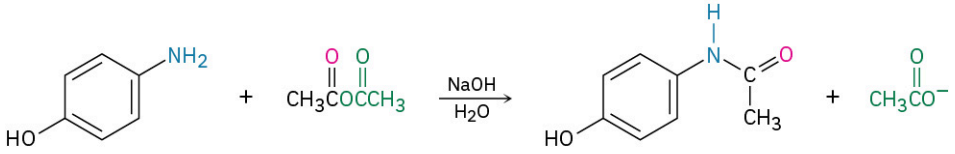
Conversion of Acid Anhydrides into Esters

Acetic anhydride is often used to prepare acetate esters from alcohols. For example, aspirin (acetylsalicylic acid) is prepared commercially by the acetylation of *o*-hydroxybenzoic acid (salicylic acid) with acetic anhydride.



Conversion of Acid Anhydrides into Amides

Acetic anhydride is also commonly used to prepare *N*-substituted acetamides from amines. For example, acetaminophen, a drug used in over-the-counter analgesics such as Tylenol, is prepared by reaction of *p*-hydroxyaniline with acetic anhydride. Only the more nucleophilic $-NH_2$ group reacts rather than the less nucleophilic –OH group.

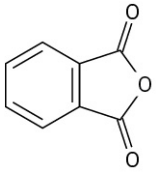


***p*-Hydroxyaniline** **Acetic anhydride** **Acetaminophen**

Notice in both of the previous reactions that only "half" of the anhydride molecule is used, while the other half acts as the leaving group during the nucleophilic acyl substitution step and produces acetate ion as a by-product. Thus, anhydrides are inefficient, and acid chlorides are normally preferred for introducing acyl substituents other than acetyl groups.

PROBLEM 21-14 Write the mechanism of the reaction between *p*-hydroxyaniline and acetic anhydride to prepare acetaminophen.

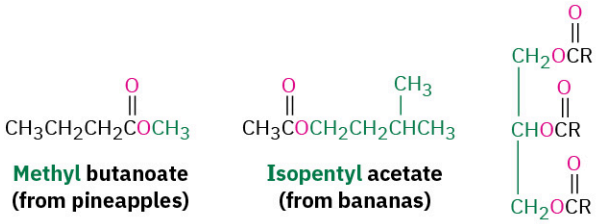
PROBLEM 21-15 What product would you expect from reaction of one equivalent of methanol with a cyclic anhydride, such as phthalic anhydride (1,2-benzenedicarboxylic anhydride)? What is the fate of the second "half" of the anhydride?



Phthalic anhydride

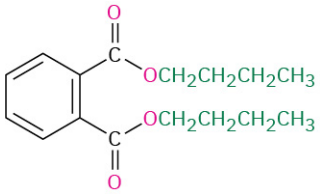
21.6 Chemistry of Esters

Esters are among the most widespread of all naturally occurring compounds. Many simple esters are pleasant-smelling liquids that are responsible for the fragrant odors of fruits and flowers. For example, methyl butanoate is found in pineapple oil, and isopentyl acetate is a constituent of banana oil. The ester linkage is also present in animal fats and in many biologically important molecules.



Methyl butanoate (from pineapples) **Isopentyl acetate (from bananas)** **A fat (R = C_{11-17} chains)**

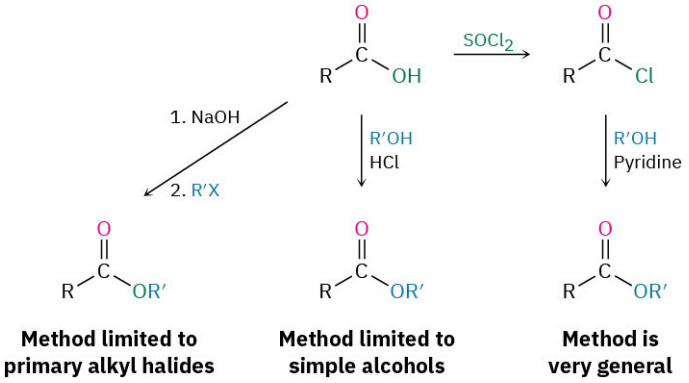
The chemical industry uses esters for a variety of purposes. Ethyl acetate, for instance, is a commonly used solvent, and dialkyl phthalates are used as plasticizers to keep polymers from becoming brittle. You may be aware that there is current concern about the possible toxicity of phthalates at high concentrations, although a recent assessment by the U.S. Food and Drug Administration found the risk to be minimal for most people, with the possible exception of male infants.



Dibutyl phthalate (a plasticizer)

Preparation of Esters

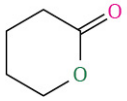
Esters are usually prepared from carboxylic acids by the methods already discussed. Thus, carboxylic acids are converted directly into esters by S_N2 reaction of a carboxylate ion with a primary alkyl halide or by Fischer esterification of a carboxylic acid with an alcohol in the presence of a mineral acid catalyst. In addition, acid chlorides are converted into esters by treatment with an alcohol in the presence of base (**Section 21.4**).



Method limited to primary alkyl halides | **Method limited to simple alcohols** | **Method is very general**

Reactions of Esters

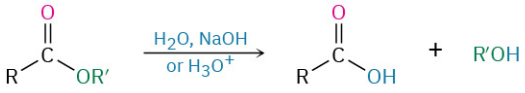
Esters undergo the same kinds of reactions that we've seen for other carboxylic acid derivatives, but they are less reactive toward nucleophiles than either acid chlorides or anhydrides. All their reactions are applicable to both acyclic and cyclic esters, called **lactones**.



A lactone (cyclic ester)

Conversion of Esters into Carboxylic Acids: Hydrolysis

An ester is hydrolyzed, either by aqueous base or aqueous acid, to yield a carboxylic acid plus an alcohol.



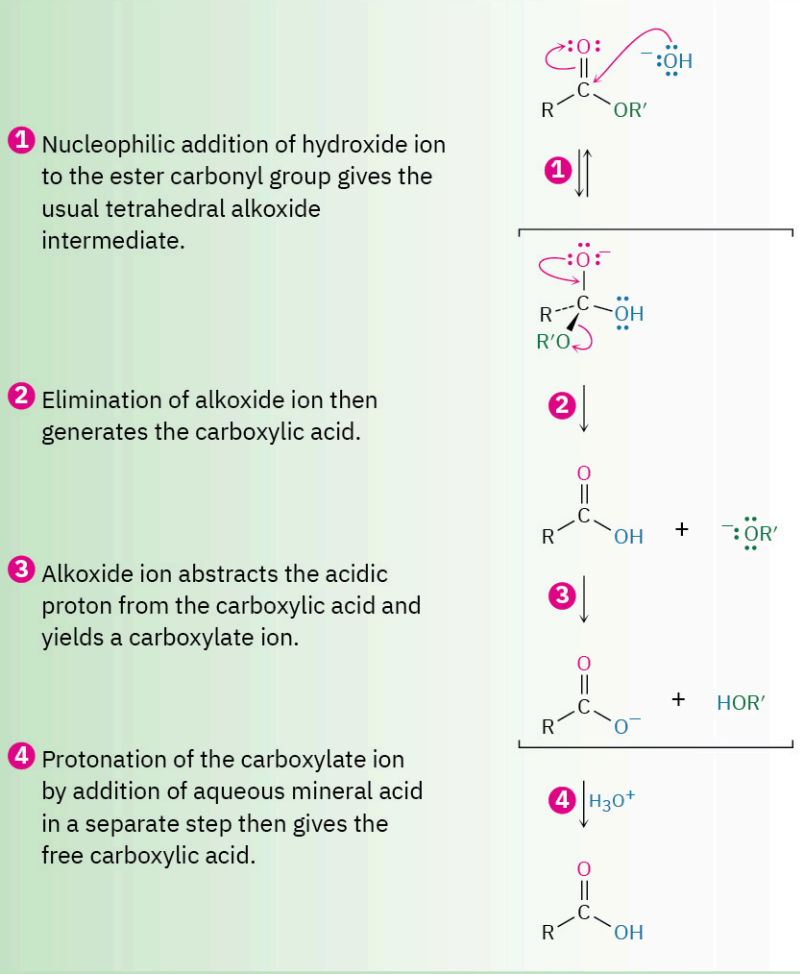
Ester **Acid** **Alcohol**

Ester hydrolysis in basic solution is called **saponification**, after the Latin word *sapo,* meaning "soap." We'll see in **Section 27.2** that soap is in fact made by boiling animal fat with aqueous base to hydrolyze the ester linkages.

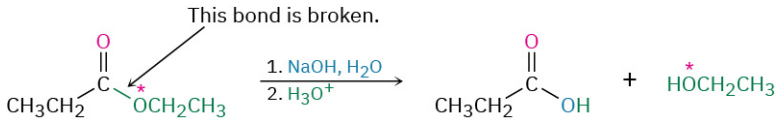
As shown in **FIGURE 21.8**, ester hydrolysis occurs through a typical nucleophilic acyl substitution pathway in which hydroxide ion is the nucleophile that adds to the ester carbonyl group to give a tetrahedral intermediate. Loss of alkoxide ion then gives a carboxylic acid, which is deprotonated to give the carboxylate ion. Addition of aqueous HCl, in a separate step after the saponification is complete, protonates the carboxylate ion and gives the carboxylic acid.

FIGURE 21.8 MECHANISM

Mechanism of base-induced ester hydrolysis (saponification).



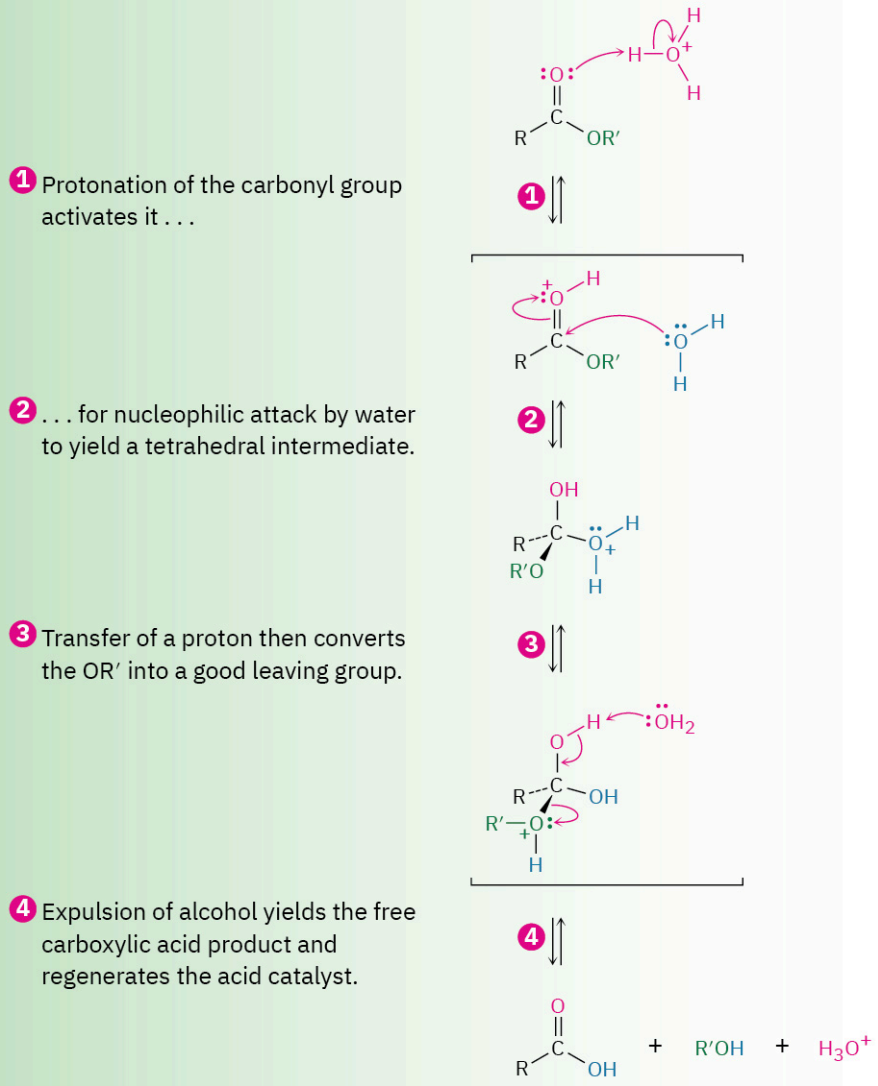
The mechanism shown in **FIGURE 21.8** is supported by isotope-labeling studies. When ethyl propanoate labeled with ^{18}O in the ether-like oxygen is hydrolyzed in aqueous NaOH, the ^{18}O label shows up exclusively in the ethanol product. None of the label remains with the propanoic acid, indicating that saponification occurs by cleavage of the C–OR′ bond rather than the CO–R′ bond.



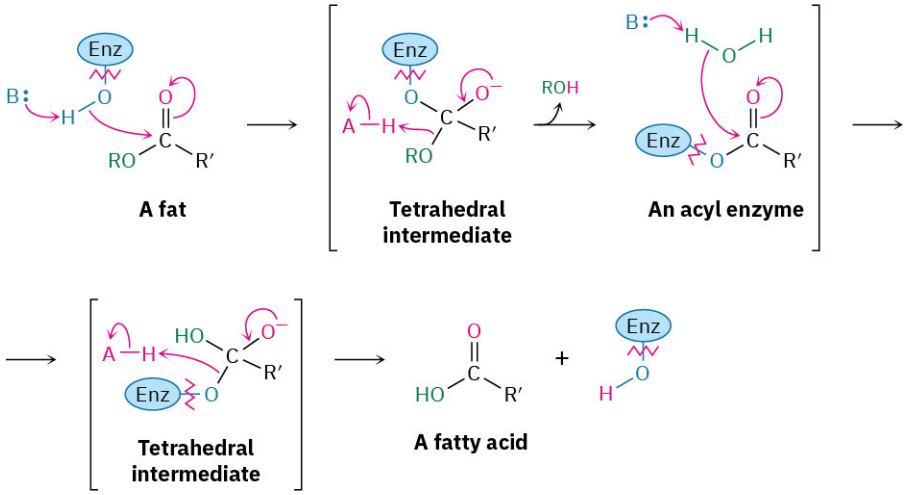
Acid-catalyzed ester hydrolysis can occur by more than one mechanism, depending on the structure of the ester. The usual pathway, however, is just the reverse of a Fischer esterification reaction (**Section 21.3**). As shown in **FIGURE 21.9**, the ester is first activated toward nucleophilic attack by protonation of the carboxyl oxygen atom, and nucleophilic addition of water then occurs. Transfer of a proton and elimination of alcohol yields the carboxylic acid. Because this hydrolysis reaction is the reverse of a Fischer esterification reaction, **FIGURE 21.9** is the reverse of **FIGURE 21.5**.

FIGURE 21.9 MECHANISM

Mechanism of acid-catalyzed ester hydrolysis. The forward reaction is a hydrolysis; the back-reaction is a Fischer esterification and is thus the reverse of FIGURE 21.5.



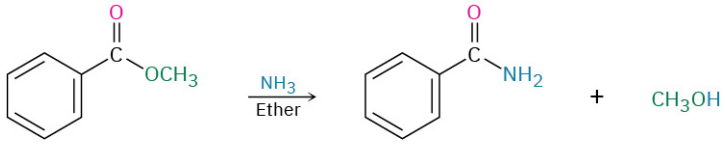
Ester hydrolysis is common in biological chemistry, particularly in the digestion of dietary fats and oils. We'll save a complete discussion of the mechanistic details of fat hydrolysis until **Section 29.2** but will note for now that the reaction is catalyzed by various lipase enzymes and involves two sequential nucleophilic acyl substitution reactions. The first is a *transesterification* reaction in which an alcohol group on the lipase adds to an ester linkage in the fat molecule to give a tetrahedral intermediate that expels alcohol and forms an acyl enzyme intermediate. The second is an addition of water to the acyl enzyme, followed by expulsion of the enzyme to give a hydrolyzed acid and a regenerated enzyme.



PROBLEM 21-16 Why is the saponification of an ester irreversible? In other words, why doesn't treatment of a carboxylic acid with an alkoxide ion yield an ester?

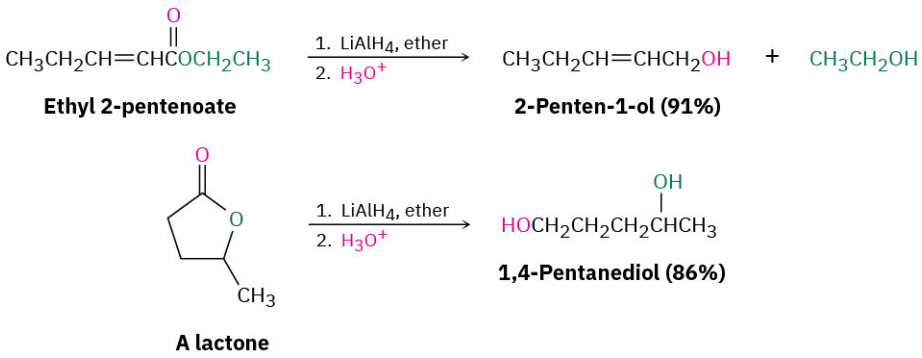
Conversion of Esters into Amides: Aminolysis

Esters react with ammonia and amines to yield amides. The reaction is not often used, however, because it's usually easier to prepare an amide by starting with an acid chloride (**Section 21.4**).

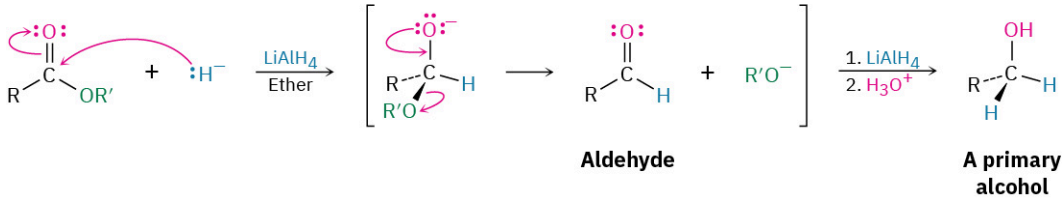


Conversion of Esters into Alcohols: Reduction

Esters are easily reduced by treatment with $LiAlH_4$ to yield primary alcohols (**Section 17.4**).

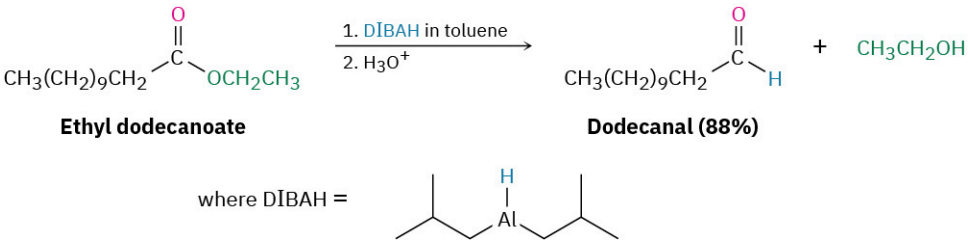


The mechanism of ester reduction is similar to that of acid chloride reduction in that a hydride ion first adds to the carbonyl group, followed by elimination of alkoxide ion to yield an aldehyde. Further reduction of the aldehyde gives the primary alcohol.

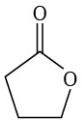


The aldehyde intermediate can be isolated if 1 equivalent of diisobutylaluminum hydride (DIBAH, or DIBAL-H) is used as the reducing agent instead of $LiAlH_4$. The reaction has to be carried out at −78 °C to avoid further reduction to the alcohol. Such partial reductions of carboxylic acid derivatives to aldehydes also occur

in numerous biological pathways, although the substrate is either a thioester or acyl phosphate rather than an ester.



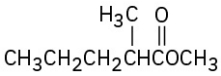
PROBLEM 21-17 What product would you expect from the reaction of butyrolactone with $LiAlH_4$? With DIBAH?



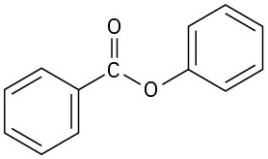
Butyrolactone

PROBLEM 21-18 Show the products you would obtain by reduction of the following esters with $LiAlH_4$:

(a)

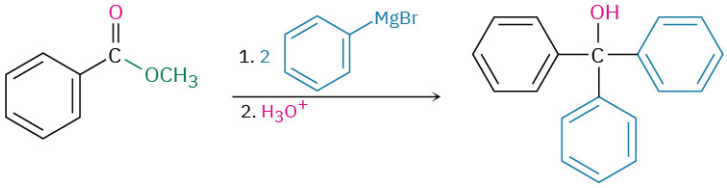


(b)



Conversion of Esters into Alcohols: Grignard Reaction

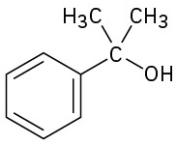
Esters react with 2 equivalents of a Grignard reagent to yield a tertiary alcohol in which two of the substituents are identical (**Section 17.5**). The reaction occurs by the usual nucleophilic substitution mechanism to give an intermediate ketone, which reacts further with the Grignard reagent to yield a tertiary alcohol.



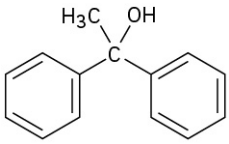
Methyl benzoate **Triphenylmethanol (96%)**

PROBLEM 21-19 What ester and what Grignard reagent might you start with to prepare the following alcohols?

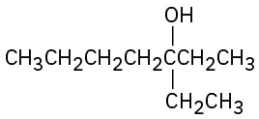
(a)



(b)

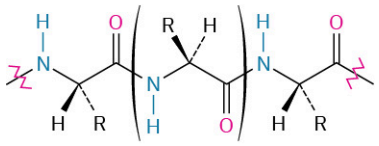


(c)

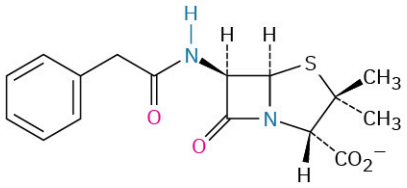


21.7 Chemistry of Amides

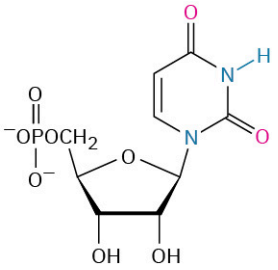
Amides, like esters, are abundant in all living organisms. Proteins, nucleic acids, and many pharmaceutical agents have amide functional groups. The reason for this abundance of amides is that they are stable in the aqueous conditions found in living organisms. Amides are the least reactive of the common acid derivatives and undergo relatively few nucleophilic acyl substitution reactions.



A protein segment



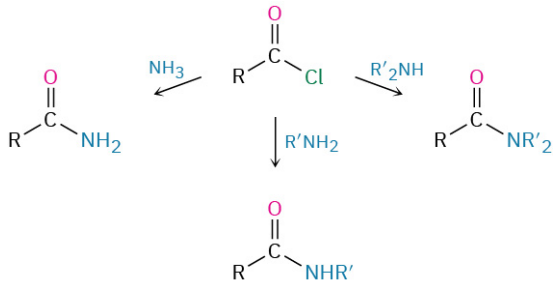
Benzylpenicillin (penicillin G)



Uridine 5′-phosphate (a ribonucleotide)

Preparation of Amides

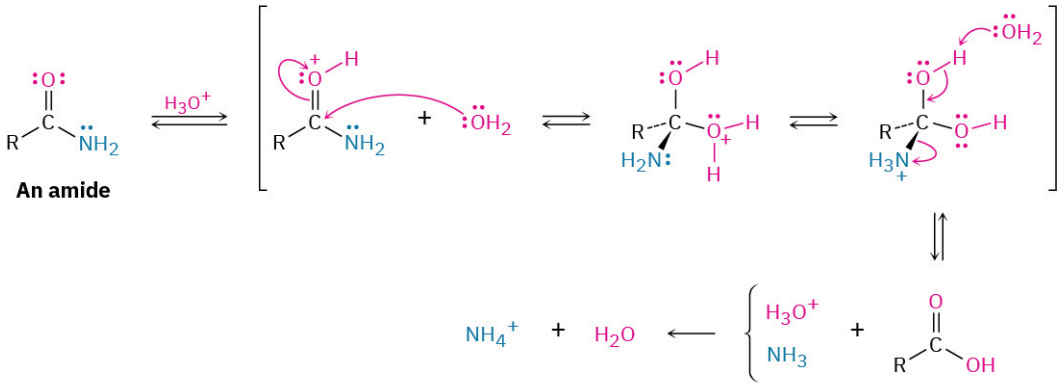
Amides are usually prepared by reaction of an amine with an acid chloride (**Section 21.4**). Ammonia, monosubstituted amines, and disubstituted amines all undergo the reaction.



Reactions of Amides

Conversion of Amides into Carboxylic Acids: Hydrolysis

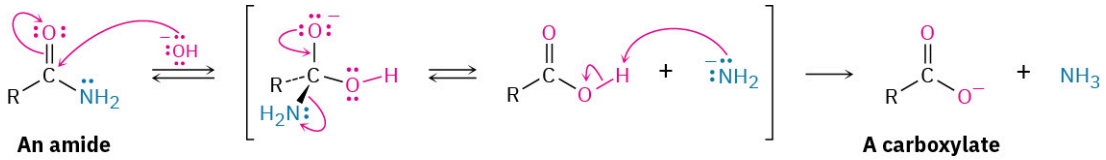
Amides undergo hydrolysis to yield carboxylic acids plus ammonia or an amine upon heating in either aqueous acid or aqueous base. The conditions required for amide hydrolysis are more extreme than those required for the hydrolysis of acid chlorides or esters, but the mechanisms are similar. Acidic hydrolysis reaction occurs by nucleophilic addition of water to the protonated amide, followed by transfer of a proton from oxygen to nitrogen to make the nitrogen a better leaving group, and subsequent elimination. The steps are reversible, with the equilibrium shifted toward product by protonation of NH_3 in the final step.



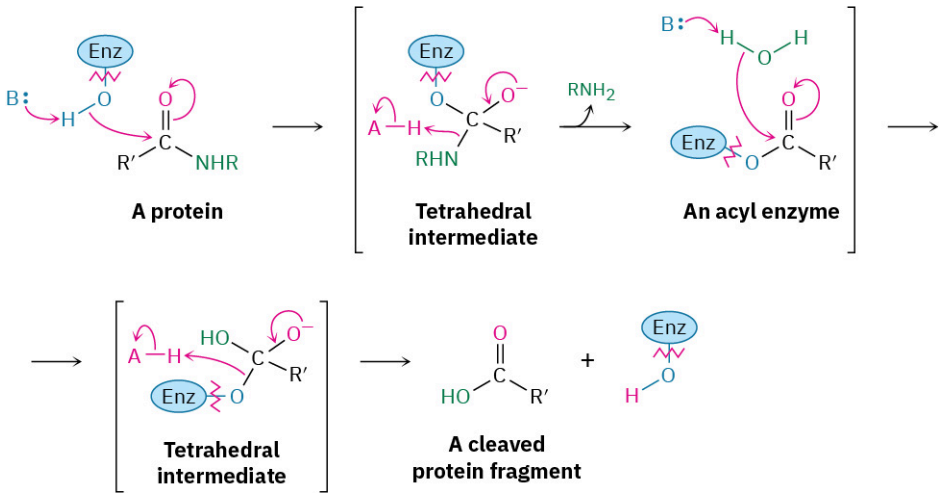
An amide

A carboxylic acid

Basic hydrolysis occurs by nucleophilic addition of OH^- to the amide carbonyl group, followed by elimination of amide ion ($^-NH_2$) and subsequent deprotonation of the initially formed carboxylic acid by ammonia. The steps are reversible, with the equilibrium shifted toward product by the final deprotonation of the carboxylic acid. Basic hydrolysis is substantially more difficult than the analogous acid-catalyzed reaction because amide ion is a very poor leaving group, making the elimination step difficult.

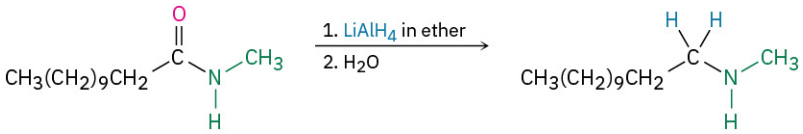


Amide hydrolysis is common in biological chemistry. Just as the hydrolysis of esters is the initial step in the digestion of dietary fats, the hydrolysis of amides is the initial step in the digestion of dietary proteins. The reaction is catalyzed by protease enzymes and occurs by a mechanism almost identical to what we just saw for fat hydrolysis. That is, an initial nucleophilic acyl substitution of an alcohol group in the enzyme on an amide linkage in the protein gives an acyl enzyme intermediate, which then undergoes hydrolysis.



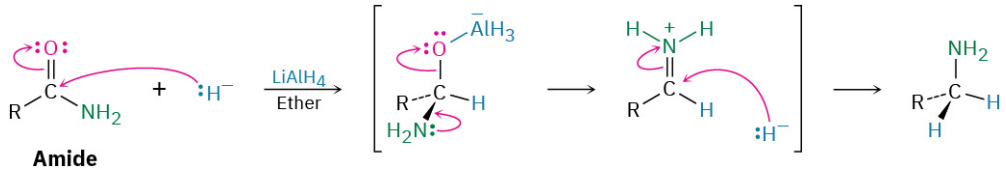
Conversion of Amides into Amines: Reduction

Like other carboxylic acid derivatives, amides can be reduced by $LiAlH_4$. The product of the reduction, however, is an amine rather than an alcohol. The net effect of an amide reduction is thus the conversion of the amide carbonyl group into a methylene group ($C{=}O \longrightarrow CH_2$). This kind of reaction is specific to amides and does not occur with other carboxylic acid derivatives.

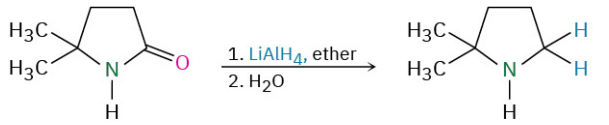


***N*-Methyldodecanamide** **Dodecylmethylamine (95%)**

Amide reduction occurs by nucleophilic addition of hydride ion to the amide carbonyl group, followed by expulsion of the *oxygen* atom as an aluminate anion leaving group to give an iminium ion intermediate. The intermediate iminium ion is further reduced by $LiAlH_4$ to yield the amine.



The reaction is effective with both acyclic and cyclic amides, or **lactams**, and is a good method for preparing cyclic amines.

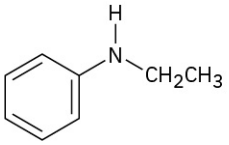


A lactam **A cyclic amine (80%)**

WORKED EXAMPLE 21.4

Synthesizing an Amine from an Amide

How could you prepare *N*-ethylaniline by reduction of an amide with $LiAlH_4$?

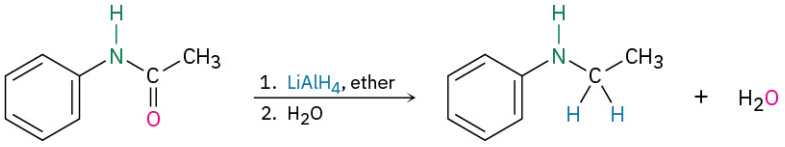


***N*-Ethylaniline**

Strategy

Reduction of an amide with $LiAlH_4$ yields an amine. To find the starting material for synthesis of *N*-ethylaniline, look for a CH_2 position next to the nitrogen atom and replace that CH_2 by C=O. In this case, the amide is *N*-phenylacetamide.

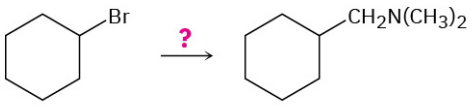
Solution



***N*-Phenylacetamide** ***N*-Ethylaniline**

PROBLEM 21-20 How would you convert *N*-ethylbenzamide to each of the following products?
(a) Benzoic acid **(b)** Benzyl alcohol **(c)** $C_6H_5CH_2NHCH_2CH_3$

PROBLEM 21-21 How would you use the reaction of an amide with $LiAlH_4$ as the key step in going from bromocyclohexane to (*N,N*-dimethylaminomethyl)cyclohexane? Write all the steps in the reaction sequence.

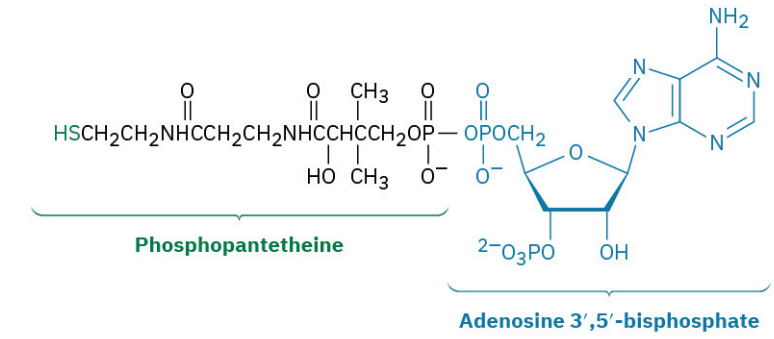


(*N,N*-Dimethylaminomethyl)cyclohexane

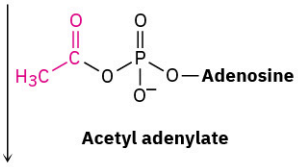
21.8 Chemistry of Thioesters and Acyl Phosphates: Biological Carboxylic Acid Derivatives

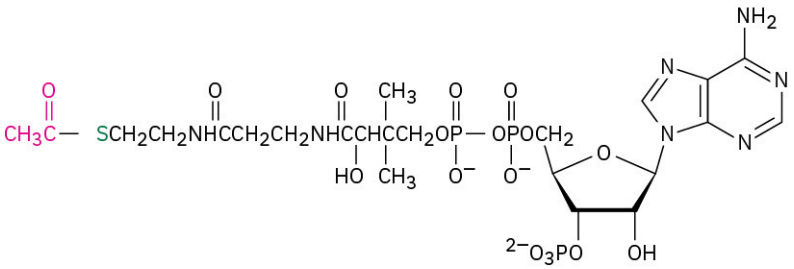
As mentioned in the chapter introduction, the substrate for a nucleophilic acyl substitution reaction in living organisms is generally either a thioester (RCOSR′) or an acyl phosphate ($RCO_2PO_3^{2-}$ or $RCO_2PO_3R'^-$). Neither is as reactive as an acid chloride or acid anhydride, yet both are stable enough to exist in living organisms while still reactive enough to undergo acyl substitution.

Acyl CoA's, such as acetyl CoA, are the most common thioesters in nature. Coenzyme A, abbreviated CoA, is a thiol formed by a phosphoric anhydride linkage (O=P–O–P=O) between phosphopantetheine and adenosine 3′,5′-bisphosphate. (The prefix *bis-* means "two" and indicates that adenosine 3′,5′-bisphosphate has two phosphate groups, one on C3′ and one on C5′.) Reaction of coenzyme A with an acyl phosphate or acyl adenylate gives acyl CoA (FIGURE 21.10). As we saw in Section 21.3 (FIGURE 21.7), formation of the acyl adenylate occurs by reaction of a carboxylic acid with ATP and is itself a nucleophilic acyl substitution reaction that takes place on phosphorus.



Coenzyme A (CoA)

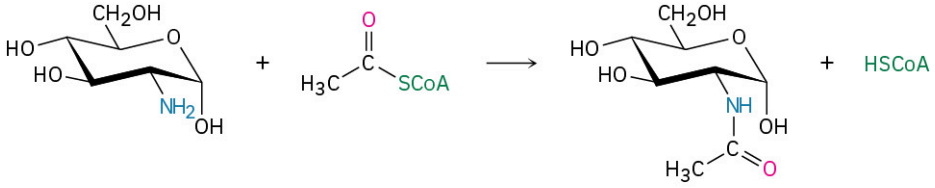




Acetyl CoA

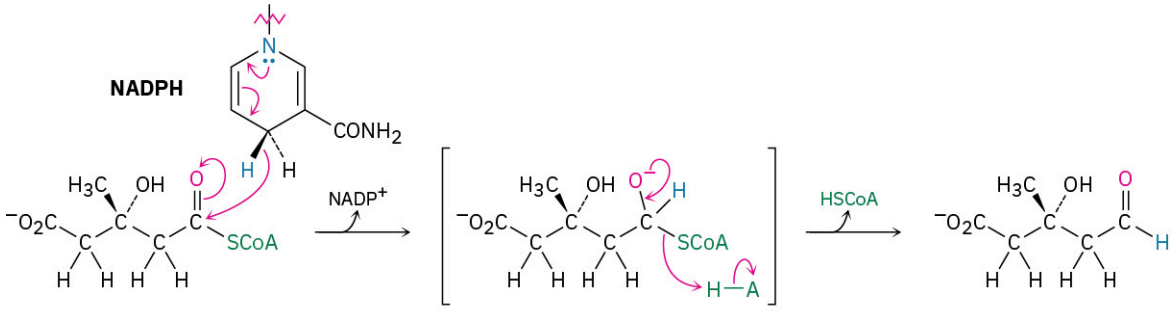
FIGURE 21.10 Formation of the thioester acetyl CoA by nucleophilic acyl substitution reaction of coenzyme A (CoA) with acetyl adenylate.

Once formed, an acyl CoA is a substrate for further nucleophilic acyl substitution reactions. For example, *N*-acetylglucosamine, a component of cartilage and other connective tissues, is synthesized by an aminolysis reaction between glucosamine and acetyl CoA.



Glucosamine (an amine) **N-Acetylglucosamine (an amide)**

Another example of a nucleophilic acyl substitution reaction on a thioester—this one a substitution by hydride ion to effect the partial reduction of a thioester to an aldehyde—occurs in the biosynthesis of mevaldehyde, an intermediate in terpenoid synthesis, which we'll discuss in some detail in **Section 27.5**. In this reaction, (3*S*)-3-hydroxy-3-methylglutaryl CoA is reduced by hydride donation from NADPH.



(3*S*)-3-Hydroxy-3-methylglutaryl CoA **(*R*)-Mevaldehyde**

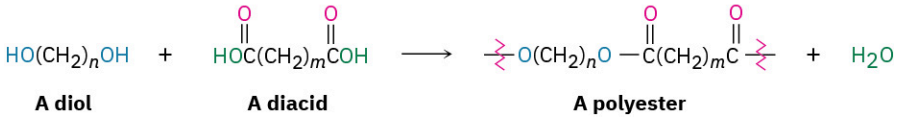
PROBLEM 21-22 Write the mechanism of the reaction shown in Figure 21.10 between coenzyme A and acetyl adenylate to give acetyl CoA.

21.9 Polyamides and Polyesters: Step-Growth Polymers

When an amine reacts with an acid chloride, an amide is formed. What would happen, though, if a *diamine* and a *diacid chloride* were allowed to react? Each partner would form two amide bonds, linking more and more molecules together until a giant polyamide resulted. In the same way, reaction of a diol with a diacid would lead to a polyester.



A diamine **A diacid chloride** **A polyamide (nylon)**



A diol **A diacid** **A polyester**

There are two main classes of synthetic polymers: *chain-growth polymers* and *step-growth polymers.* The polyethylene and other alkene and diene polymers that we saw in **Section 8.10** and **Section 14.6** are called **chain-growth polymers** because they are produced in chain-reaction processes. An initiator adds to a C=C bond to give a reactive intermediate, which adds to a second alkene molecule to produce a new intermediate, which adds to a third molecule, and so on. By contrast, polyamides and polyesters are **step-growth polymers** because each bond in the polymer is formed independently in a discrete step, often the nucleophilic acyl substitution reaction of a carboxylic acid derivative. Some commercially important step-growth polymers are shown in **TABLE 21.2**.

TABLE 21.2 Some Common Step-Growth Polymers and Their Uses

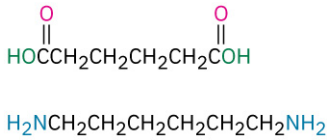
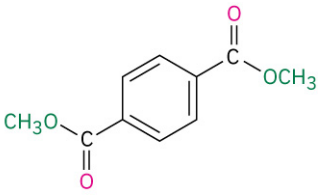
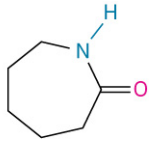
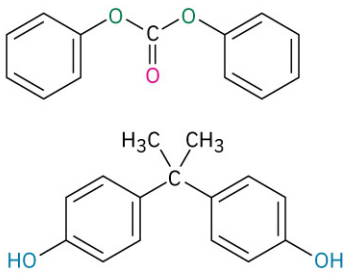
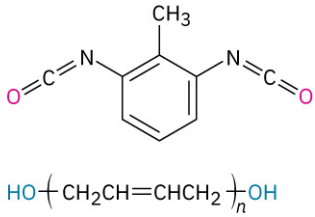
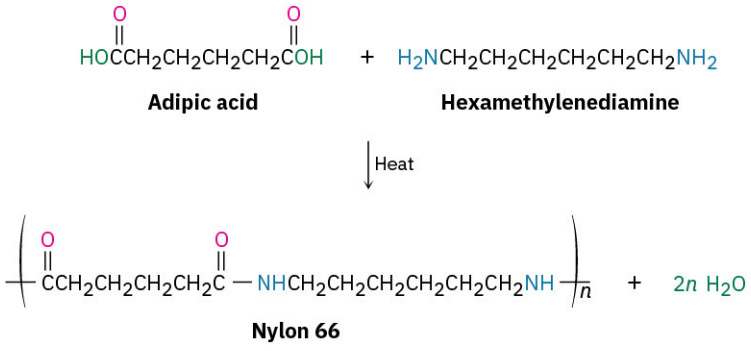
Monomers	Structure	Polymer	Uses
Adipic acid + Hexamethylenediamine		Nylon 66	Fibers, clothing, tire cord
Dimethyl terephthalate + Ethylene glycol		Dacron, Mylar, Terylene	Fibers, clothing, films, tire cord

TABLE 21.2 Some Common Step-Growth Polymers and Their Uses

Monomers	Structure	Polymer	Uses
Caprolactam		Nylon 6, Perlon	Fibers, castings
Diphenyl carbonate + Bisphenol A		Lexan, polycarbonate	Equipment housing, molded articles
Toluene-2,6-diisocyanate + Poly(2-butene-1,4-diol)		Polyurethane, Spandex	Fibers, coatings, foams

Polyamides (Nylons)

The best known step-growth polymers are the polyamides, or *nylons,* first prepared in 1930 by Wallace Carothers at the DuPont Company by heating a diamine with a diacid. For instance, nylon 66 is prepared by reaction of adipic acid (hexanedioic acid) with hexamethylenediamine (1,6-hexanediamine) at 280 °C. The designation "66" indicates the number of carbon atoms in the diamine (the first 6) and the diacid (the second 6).

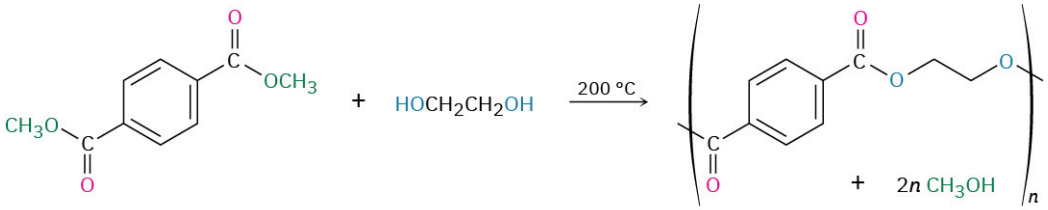


Nylon 66

Nylons are used both in engineering applications and in making fibers. A combination of high impact strength and abrasion resistance makes nylon an excellent metal substitute for bearings and gears. As fiber, nylon is used in a variety of applications, from clothing to tire cord to ropes.

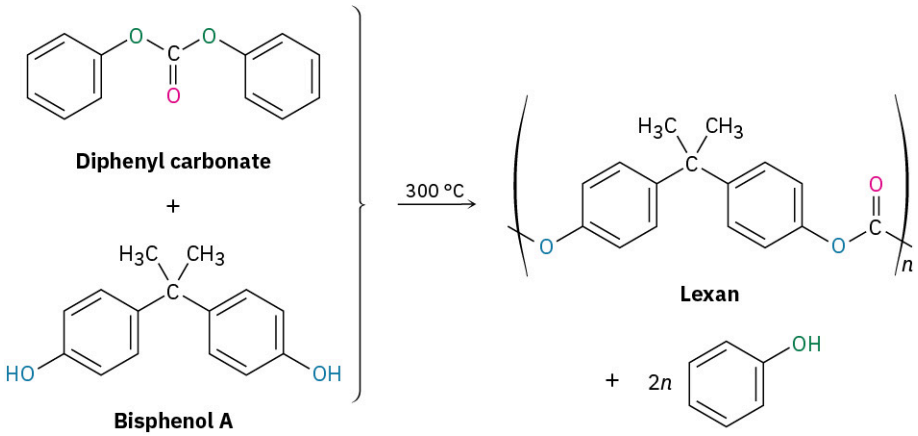
Polyesters

The most generally useful polyester is made by reaction between dimethyl terephthalate (dimethyl 1,4-benzenedicarboxylate) and ethylene glycol (1,2-ethanediol). The product is used under the trade name Dacron to make clothing fiber or tire cord and under the name Mylar to make recording tape. The tensile strength of poly(ethylene terephthalate) film is nearly equal to that of steel.



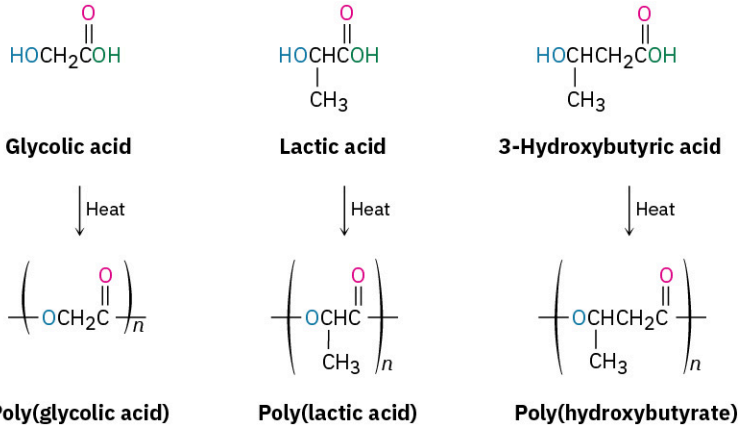
Dimethyl terephthalate **Ethylene glycol** **A polyester (Dacron, Mylar)**

Lexan, a polycarbonate prepared from diphenyl carbonate and bisphenol A, is another commercially valuable polyester. Lexan has an unusually high impact strength, making it valuable for use in bicycle helmets and laptop cases.



Sutures and Biodegradable Polymers

Because plastics are too often thrown away rather than recycled, much work has been carried out on developing biodegradable polymers, which can be broken down rapidly in landfills by soil microorganisms. Among the most common biodegradable polymers are poly(glycolic acid) (PGA), poly(lactic acid) (PLA), and poly(hydroxybutyrate) (PHB). All are polyesters and are therefore susceptible to hydrolysis of their ester links. Copolymers of PGA with PLA have found a particularly wide range of uses. A 90/10 copolymer of poly(glycolic acid) with poly(lactic acid) is used to make absorbable sutures, for instance. The sutures are entirely hydrolyzed and absorbed by the body within 90 days after surgery.



Poly(glycolic acid) **Poly(lactic acid)** **Poly(hydroxybutyrate)**

In Europe, interest has centered particularly on poly(hydroxybutyrate), which can be made into films for packaging as well as into molded items. The polymer degrades within four weeks in landfills, both by ester hydrolysis and by an E1cB elimination reaction of the oxygen atom β to the carbonyl group. The use of poly(hydroxybutyrate) is limited at present by its cost—about four times that of polypropylene.

PROBLEM 21-23 Draw structures of the step-growth polymers you would expect to obtain from the following reactions:

(a) $BrCH_2CH_2CH_2Br + HOCH_2CH_2CH_2OH \xrightarrow{\text{Base}} ?$

(b) $HOCH_2CH_2OH + HO_2C(CH_2)_6CO_2H \xrightarrow{H_2SO_4\ \text{catalyst}} ?$

(c) $H_2N(CH_2)_6NH_2 + ClC(=O)(CH_2)_4C(=O)Cl \longrightarrow ?$

PROBLEM 21-24 Kevlar, a nylon polymer prepared by reaction of 1,4-benzenedicarboxylic acid (terephthalic acid) with 1,4-benzenediamine (*p*-phenylenediamine), is so strong that it's used to make bulletproof vests. Draw the structure of a segment of Kevlar.

21.10 Spectroscopy of Carboxylic Acid Derivatives

Infrared Spectroscopy

All carbonyl-containing compounds have intense IR absorptions in the range 1650 to 1850 cm^{-1}. As shown in TABLE 21.3 the exact position of the absorption provides information about the specific kind of carbonyl group. For comparison, the IR absorptions of aldehydes, ketones, and carboxylic acids are included in the table, along with values for carboxylic acid derivatives.

TABLE 21.3 **Infrared Absorptions of Some Carbonyl Compounds**

Carbonyl type	Example	Absorption (cm^{-1})
Saturated acid chloride	Acetyl chloride	1810
Aromatic acid chloride	Benzoyl chloride	1770
Saturated acid anhydride	Acetic anhydride	1820, 1760
Saturated ester	Ethyl acetate	1735
Aromatic ester	Ethyl benzoate	1720
Saturated amide	Acetamide	1690
Aromatic amide	Benzamide	1675
N-Substituted amide	*N*-Methylacetamide	1680
N,N-Disubstituted amide	*N,N*-Dimethylacetamide	1650
Saturated aldehyde	Acetaldehyde	1730
Saturated ketone	Acetone	1715
Saturated carboxylic acid	Acetic acid	1710

Acid chlorides are easily detected by their characteristic absorption near 1810 cm^{-1}. Acid anhydrides can be identified by a pair of absorptions in the carbonyl region, one at 1820 cm^{-1} and another at 1760 cm^{-1}. Note that each of these functional groups has a strong electron-withdrawing group attached to the carbonyl. The inductive withdrawal of electron density shortens the C=O bond, thereby raising its stretching frequency. Esters are detected by their absorption at 1735 cm^{-1}, a position somewhat higher than that for either aldehydes or ketones. Amides, by contrast, absorb near the low-wavenumber end of the carbonyl region, with the degree of substitution on nitrogen affecting the exact position of the IR band. That is, for amides, the delocalization of

electron density (resonance) from nitrogen into the carbonyl lengthens the C=O bond and lowers its stretching frequency.

PROBLEM 21-25 What kinds of functional groups might compounds have if they show the following IR absorptions?
(a) Absorption at 1735 cm^{-1} **(b)** Absorption at 1810 cm^{-1}
(c) Absorptions at 2500 to 3300 cm^{-1} and 1710 cm^{-1} **(d)** Absorption at 1715 cm^{-1}

PROBLEM 21-26 Propose structures for compounds that have the following formulas and IR absorptions:
(a) $C_6H_{12}O_2$, 1735 cm^{-1} **(b)** C_4H_9NO, 1650 cm^{-1} **(c)** C_4H_5ClO, 1780 cm^{-1}

Nuclear Magnetic Resonance Spectroscopy

Hydrogens on the carbon next to a carbonyl group are slightly deshielded and absorb near 2 δ in the ^1H NMR spectrum, although the exact identity of the carbonyl group can't be determined. FIGURE 21.11 shows the ^1H NMR spectrum of ethyl acetate.

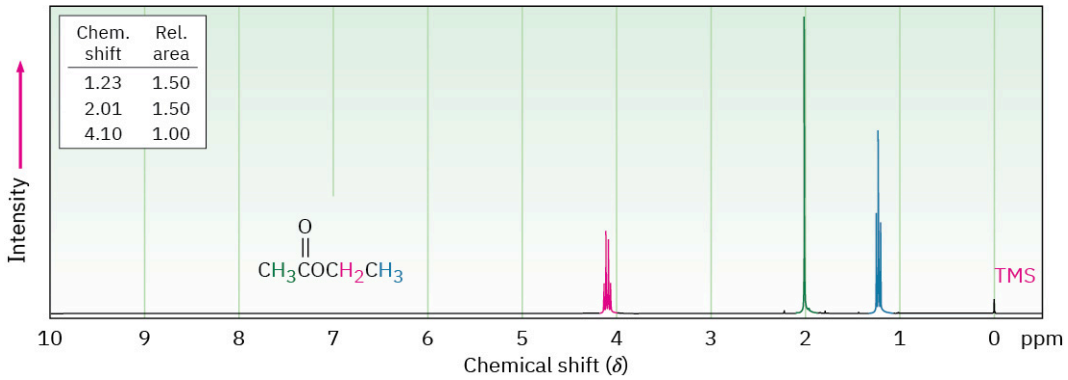


FIGURE 21.11 ^1H NMR spectrum of ethyl acetate.

Although ^{13}C NMR is useful for determining the presence or absence of a carbonyl group in a molecule, the identity of the carbonyl group is difficult to determine. Aldehydes and ketones absorb near 200 δ, while the carbonyl carbon atoms of various acid derivatives absorb in the range 160 to 180 δ (TABLE 21.4).

TABLE 21.4 ^{13}C NMR Absorptions of Some Carbonyl Compounds

Compound	Absorption (δ)
Acetic acid	177.3
Ethyl acetate	170.7
Acetyl chloride	170.3
Acetamide	172.6
Acetic anhydride	166.9
Acetone	205.6
Acetaldehyde	201.0

CHEMISTRY MATTERS

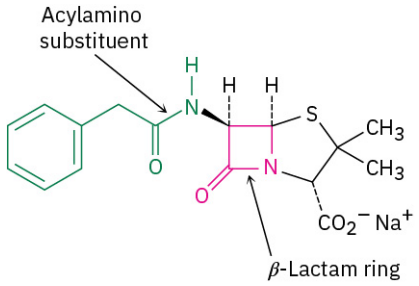
β-Lactam Antibiotics

You should never underestimate the value of hard work and logical thinking, but it's also true that blind luck often plays a role in most real scientific breakthroughs. What has been called "the supreme example of luck in all scientific history" occurred in the late summer of 1928, when the Scottish bacteriologist Alexander Fleming went on vacation, leaving in his lab a culture plate recently inoculated with the bacterium *Staphylococcus aureus*.

While Fleming was away, an extraordinary chain of events occurred. First, a nine-day cold spell lowered the laboratory temperature to a point where the *Staphylococcus* in the culture plate could not grow. During this time, spores from a colony of the mold *Penicillium notatum,* being grown in a lab on the floor below, wafted up into Fleming's lab and landed in the culture plate. The temperature then rose, and both *Staphylococcus* and *Penicillium* began to grow. On returning from vacation, Fleming discarded the plate into a tray of antiseptic, intending to sterilize it. Evidently, though, the plate did not sink deeply enough into the antiseptic, because when Fleming happened to glance at it a few days later, what he saw changed the course of history. He noticed that the growing *Penicillium* mold appeared to dissolve the colonies of staphylococci.

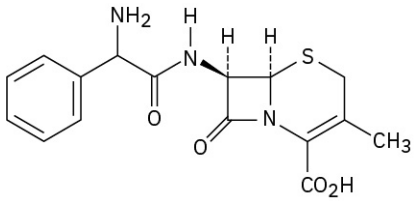
Fleming realized that the *Penicillium* mold must be producing a chemical that killed the *Staphylococcus* bacteria, and he spent several years trying to isolate the substance. Finally, in 1939, the Australian pathologist Howard Florey and the German refugee Ernst Chain managed to isolate the active substance, called *penicillin.* The dramatic ability of penicillin to cure infections in mice was soon demonstrated, and successful tests in humans followed shortly thereafter. By 1943, penicillin was being produced on a large scale for military use in World War II, and by 1944 it was being used on civilians. Fleming, Florey, and Chain shared the 1945 Nobel Prize in Physiology or Medicine.

Now called benzylpenicillin, or penicillin G, the substance first discovered by Fleming is but one member of a large class of so-called ***β*-lactam antibiotics**, compounds with a four-membered lactam (cyclic amide) ring. The four-membered lactam ring is fused to a five-membered, sulfur-containing ring, and the carbon atom next to the lactam carbonyl group is bonded to an acylamino substituent, RCONH–. This acylamino side chain can be varied in the laboratory to provide many hundreds of penicillin analogs with different biological activity profiles. Ampicillin, for instance, has an α-aminophenylacetamido substituent [$PhCH(NH_2)CONH–$].



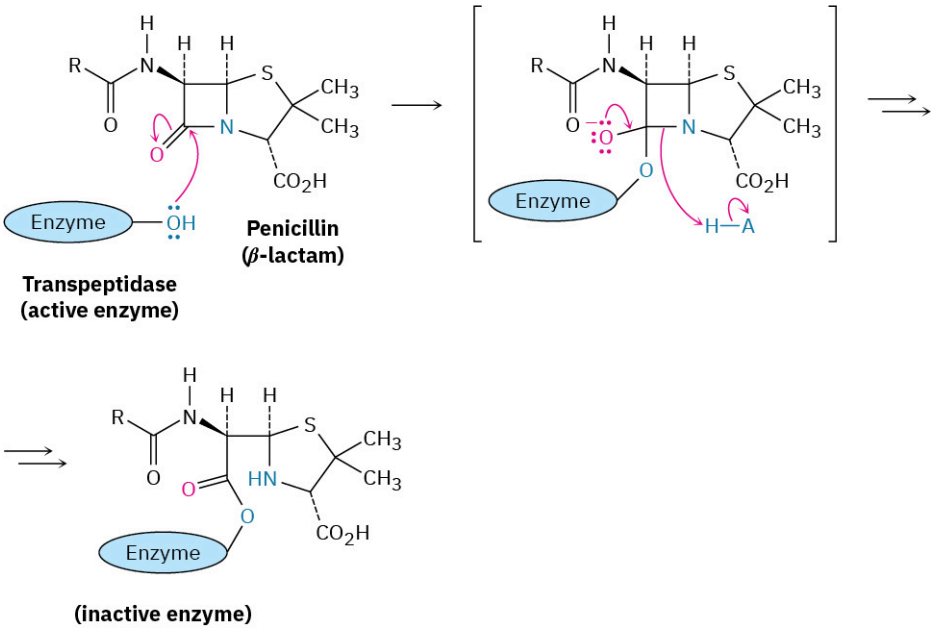
Benzylpenicillin (penicillin G)

Closely related to the penicillins are the *cephalosporins*, a group of *β*-lactam antibiotics that contain an unsaturated, six-membered, sulfur-containing ring. Cephalexin, marketed under the trade name Keflex, is an example. Cephalosporins generally have much greater antibacterial activity than penicillins, particularly against resistant strains of bacteria.



Cephalexin (a cephalosporin)

The biological activity of penicillins and cephalosporins is due to the presence of the strained β-lactam ring, which reacts with and deactivates the transpeptidase enzyme needed to synthesize and repair bacterial cell walls. With the wall either incomplete or weakened, the bacterial cell ruptures and dies.



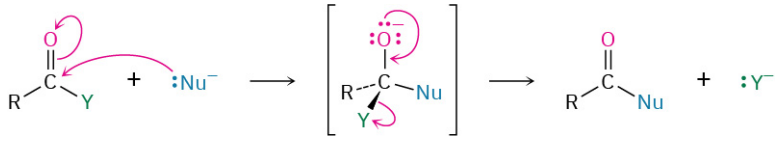
Key Terms

- **acid anhydride**
- **acid halide**
- **acyl phosphate**
- **amide**
- **carboxylic acid derivative**
- **chain-growth polymer**
- **ester**
- **Fischer esterification reaction**
- **lactam**
- **lactone**
- **nucleophilic acyl substitution reaction**
- **saponification**
- **step-growth polymer**
- **thioester**

Summary

Carboxylic acid derivatives—compounds in which the –OH group of a carboxylic acid has been replaced by another substituent—are among the most widely occurring of all molecules and are involved in almost all biological pathways. In this chapter, we covered the chemistry necessary for understanding them and thus also necessary for understanding living organisms. **Acid halides, acid anhydrides, esters,** and **amides** are the most common such derivatives in the laboratory; **thioesters** and **acyl phosphates** are common in biological molecules.

The chemistry of carboxylic acid derivatives is dominated by the **nucleophilic acyl substitution reaction**. Mechanistically, these substitutions take place by addition of a nucleophile to the polar carbonyl group of the acid derivative to give a tetrahedral intermediate, followed by expulsion of a leaving group.



The reactivity of an acid derivative toward substitution depends both on the steric environment near the carbonyl group and on the electronic nature of the substituent, Y. The reactivity order is acid halide > acid anhydride > thioester > ester > amide.

The most common reactions of carboxylic acid derivatives are substitution by water to yield an acid (hydrolysis), by an alcohol to yield an ester (alcoholysis), by an amine to yield an amide (aminolysis), by hydride ion to yield

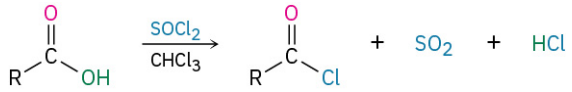
an alcohol (reduction), and by an organomagnesium halide to yield an alcohol (Grignard reaction).

Step-growth polymers, such as polyamides and polyesters, are prepared by reactions between difunctional molecules. Polyamides (nylons) are formed by reaction between a diacid and a diamine; polyesters are formed from a diacid and a diol.

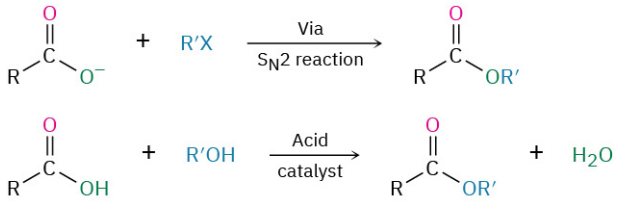
IR spectroscopy is a valuable tool for the structural analysis of acid derivatives. Acid chlorides, anhydrides, esters, and amides all show characteristic IR absorptions that can be used to identify these functional groups.

Summary of Reactions

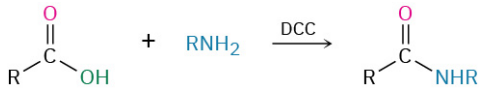
1. Reactions of carboxylic acids (**Section 21.3**)
   a. Conversion into acid chlorides



   b. Conversion into esters



   c. Conversion into amides



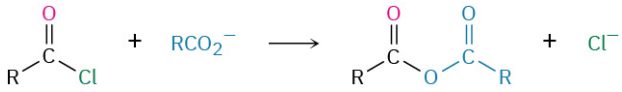
   d. Reduction to yield primary alcohols



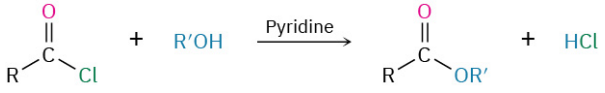
2. Reactions of acid chlorides (**Section 21.4**)
   a. Hydrolysis to yield acids



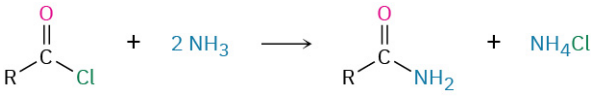
   b. Reaction with carboxylates to yield anhydrides



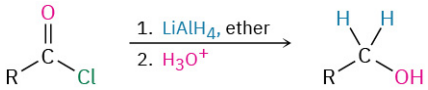
   c. Alcoholysis to yield esters



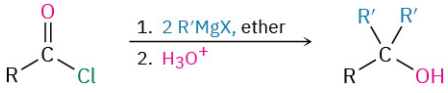
   d. Aminolysis to yield amides



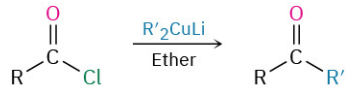
   e. Reduction to yield primary alcohols



   f. Grignard reaction to yield tertiary alcohols



g. Diorganocopper reaction to yield ketones



3. Reactions of acid anhydrides (**Section 21.5**)

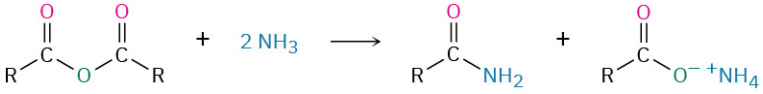
a. Hydrolysis to yields acids



b. Alcoholysis to yield esters

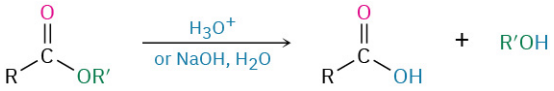


c. Aminolysis to yield amides

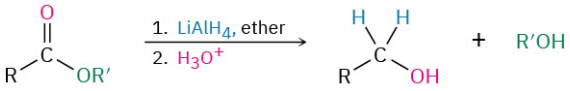


4. Reactions of esters (**Section 21.6**)

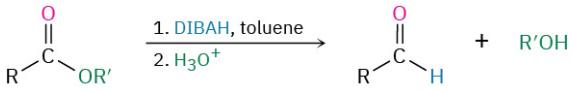
a. Hydrolysis to yields acids



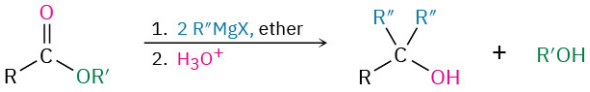
b. Reduction to yield primary alcohols



c. Partial reduction to yield aldehydes

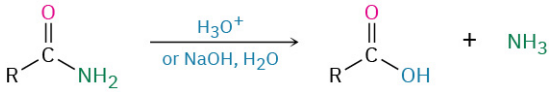


d. Grignard reaction to yield tertiary alcohols

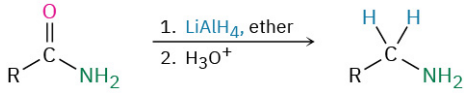


5. Reactions of amides (**Section 21.7**)

a. Hydrolysis to yields acids



b. Reduction to yield amines



Additional Problems

Visualizing Chemistry

PROBLEM Name the following compounds:

21-27